MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

DECODER, BCD TO DECIMAL, DIGITAL, KEY CODE BMH

This specification is approved for use by the Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document establishes the requirements and test procedures for the acquisition of an electronic BCD to decimal decoder module.

1.2 Classification. This specification covers two classes of modules.

Class I and class II as specified in MIL-STD-1389.

2. APPLICABLE DOCUMENTS

2.1 Government specifications and standards. Unless otherwise specified, the following specifications and standards, of the issue listed in that issue of the Department of Defense Index of Specifications and Standards specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-28787 - Modules, Standard Electronic, General Specification for.

STANDARDS

MILITARY

MIL-STD-1389 - Design Requirements for Standard Electronic Modules.
MIL-STD-1665 - Test Equipment for the Standard Electronic Modules Program.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 General requirements. The module shall comply with the requirements of MIL-M-28787 except as modified herein.

3.2 Correlation requirements. Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787.

* 3.3 Failure rate requirements. All modules acquired to this specification shall have a predicted failure rate no greater than 0.637 failures per million operating hours for class I or 1.941 failures per million operating hours for class II, when predicted in accordance with MIL-STD-1389. When a class II module is used in a class I environment, failure rate for class I shall be used for reliability prediction (see 6.3).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Naval Electronic Systems Command, ATTN: ELEX 8111, Department of the Navy, Washington, D.C. 20360, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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* 3.4 Performance requirements. The module shall conform to the electrical requirements specified in table I. Class I modules shall conform to the electrical requirements of class I only, and class II modules shall conform to the electrical requirements of class I and class II (see 6.3).

3.5 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

3.5.1 Key code. The module key code shall be BMH.

3.5.2 Size. The module size shall be 1A.

3.5.3 Weight. The module shall have a maximum weight of 35 grams.

3.5.4 Contact pins. The module shall have 40 contact pins positioned A1 through A40.

3.5.5 Fin position. The module shall have an insertion and extraction and heat dissipation fin on increment A.

3.5.6 Depth. The module shall have no additional depth requirements.

3.5.7 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes.

3.5.8 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part number M28787/258-1 for class I and M28787/258-2 for class II, except no marking shall interfere with heat dissipation from the fin (for example, affixing a label which may be a heat barrier).

3.6 Circuit pin configuration. The function diagram and pin assignments shall conform to figure 1.

3.7 Acceptance test. All production modules shall be tested in accordance with footnotes 1 and 2 of table I.

3.8 End-of-life limits. Modules which have been plugged into an operating system, or exceed 2 years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits of table I (see 6.2).

3.9 Power supply voltage. The module shall be capable of conforming to the requirements of table I for a power supply voltage (V_{CC}) of 3.0 to 18 V.

4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply.

4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

4.3 Test equipment. Test equipment specified on figures and in tables with a type designator shall be as specified in MIL-STD-1665. Other equipment and components shall be as specified on the figures. All resistors shall be metal film and all capacitors shall be ceramic unless otherwise specified.

4.4 Test procedures. The module shall be tested as specified in table I.

* 4.4.1 Alternate test methods. Alternate test methods may be employed if approved by the Standard Electronic Modules Program Quality Assurance Activity and they conform to the requirement of MIL-M-28787.

* 4.4.2 Order of testing. Test conditions A through EE shall be performed in any order with the following exceptions. If test conditions A, C, or D are part of a sequence of tests, the order of testing shall be as follows: A, C, D where applicable, followed by the remaining tests in the sequence.

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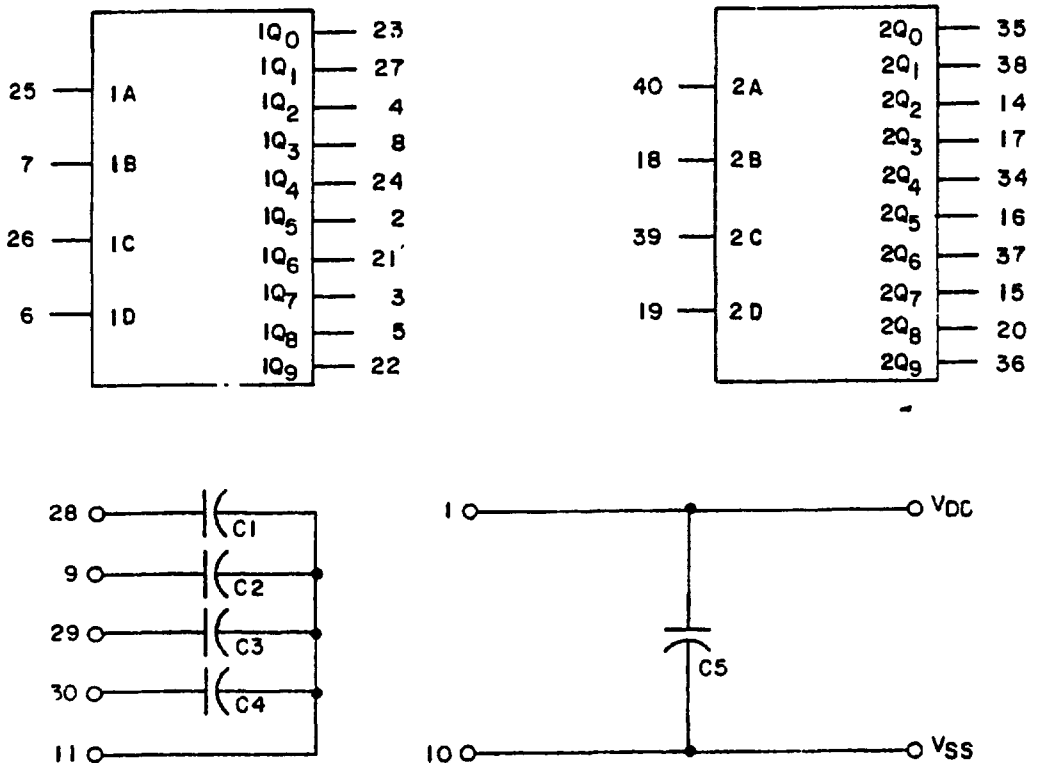


*FIGURE 1. BMH module function diagram, BCD to decimal decoder.

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* TABLE I. Module electrical requirements.

Test condition	Parameter	Ref test table	Test type 1/-3/	25° ±5°C 4/		Class I 4/				Class II 4/				Unit
						0° ±5°C		60° ±5°C		-55° ±5°C		100° ±5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
A1-A6	Isolation													
	Initial	III	S	10	∞	10	∞	10	∞	10	∞	10	∞	MΩ
	End-of-life	III	S	1.0	∞	1.0	∞	1.0	∞	1.0	∞	1.0	∞	MΩ
C1-C8	Negative input clamp diode voltage	IV	X	-0.4	-6.0	-0.4	-6.0	-0.4	-6.0	-0.4	-6.0	-0.4	-6.0	V
D1-D8	Positive input clamp diode voltage	V	X	0.4	1.5	0.4	1.5	0.4	1.5	0.4	1.5	0.4	1.5	V
E1-E10	Functional	VI	X	← See table VI →										
F1-F20	Static output voltage, low level	VII	X	0.0	0.05	0.0	0.05	0.0	0.05	0.0	0.05	0.0	0.05	V
G1-G20	Static output voltage, high level	VIII	X	4.95	5.05	4.95	5.05	4.95	5.05	4.95	5.05	4.95	5.05	V
H1-H20	Static output voltage, low level	IX	X	0.0	0.4	0.0	0.4	0.0	0.4	0.0	0.4	0.0	0.4	V
J1-J20	Static output voltage, high level	X	X	4.6	5.0	4.6	5.0	4.6	5.0	4.6	5.0	4.6	5.0	V
K1-K20	Static output voltage, low level	XI	X	0.0	1.5	0.0	1.5	0.0	1.5	0.0	1.5	0.0	1.5	V
L1-L20	Static output voltage, high level	XII	X	13.5	15.0	13.5	15.0	13.5	15.0	13.5	15.0	13.5	15.0	V
M1-M8	Input current, high level	XIII	X	0	200	---	---	0	200	---	---	0	200	nA
N1-N8	Input current, low level	XIV	X	0	-200	---	---	0	-200	---	---	0	-200	nA
P	Static power supply current	XV	X	0	40	0	40	0	580	0	40	0	1,200	μA
Q1-Q20	Transition time (t_{TLH})	XVI	S	35	350	35	350	38	385	26	265	43	430	ns
Q21-Q40	Transition time (t_{TLH})	XVI	Q	83	830	83	830	91	915	63	630	101	1,015	ns

See footnotes at end of table.

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* TABLE I. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/-3/	25° ±5°C 4/		Class I 4/				Class II 4/				Unit
						0° ±5°C		60° ±5°C		-55° ±5°C		100° ±5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
R1-R20	Transition time (t_{THL})	XVII	S	17	175	17	175	19	195	13	135	21	215	ns
R21-R40	Transition time (t_{THL})	XVII	Q	65	655	65	655	72	725	50	500	80	805	ns
S1-S60	Propagation delay time (t_{PLH})	XVIII	S	48	480	48	480	53	530	36	365	59	590	ns
S61-S120	Propagation delay time (t_{PLH})	XVIII	Q	68	680	68	680	75	750	51	515	83	835	ns
T1-T60	Propagation delay time (t_{PHL})	XIX	S	48	480	48	480	53	530	36	365	59	590	ns
T61-T120	Propagation delay time (t_{PHL})	XIX	Q	68	680	68	680	75	750	51	515	83	835	ns
U1-U20	Transition time (t_{TLH})	XX	Q	15	150	15	150	16	165	11	115	18	185	ns
V1-V20	Transition time (t_{THL})	XXI	Q	7	75	7	75	8	85	5	55	9	90	ns
W1-W60	Propagation delay time (t_{PLH})	XXII	Q	20	200	20	200	22	220	15	150	24	245	ns
X1-X60	Propagation delay time (t_{PHL})	XXIII	Q	20	200	20	200	22	220	15	150	24	245	ns
Y1-Y20	Transition time (t_{TLH})	XXIV	Q	11	110	11	110	12	120	8	85	13	135	ns
Z1-Z20	Transition time (t_{THL})	XXV	Q	5	55	5	55	6	60	4	42	6	67	ns
AA1-AA60	Propagation delay time (t_{PLH})	XXVI	Q	14	140	14	140	15	155	10	105	17	170	ns
BB1-BB60	Propagation delay time (t_{PHL})	XXVII	Q	14	140	14	140	15	155	10	105	17	170	ns
CC1-CC4	Capacitor	XXVIII	X	176	264	176	264	176	264	176	264	176	264	pF
DD	Power supply filter capacitance	XXIX	S	0.007	0.013	0.007	0.013	0.007	0.013	0.007	0.013	0.007	0.013	µF
EE1-EE8	Input capacitance	XXX	Q	2	15	2	15	2	15	2	15	2	15	pF

1/ "X" parameters shall be 100 percent tested during module acceptance tests.

2/ "S" parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot, whichever is greater) at 25°C.

3/ "Q" parameters shall be tested during qualification testing only.

4/ Unless otherwise specified limits are for initial and end-of-life.

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4.5 Qualification testing. Initial qualification tests and periodic-check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions.

4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A through EE in table I shall be performed in the order specified in 4.4.2 using the initial limits at 25° ±5°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test conditions C through L, and P through EE shall be performed in the order specified in 4.4.2 using the initial limits for applicable operating temperature. Test conditions M and N shall be performed at 25°C, 60°C, and 100°C only, using the initial limits for the applicable operating temperature. At all other times during the operating temperature tests, the module shall be operated in the test circuit shown on figure 2.

4.5.1.1 Transient thermal. For the transient thermal test, use the same test conditions as the maximum class temperature conditions.

4.5.1.2 Component temperature test. The component temperature test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 3.

4.5.2 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787. During the vibration tests, the module shall be operated in the test circuit shown on figure 4. Following the vibration tests, the electrical tests of table I which are marked with an "X" in the test type column, and test condition DD shall be performed using the initial limits at 25° ±5°C ambient.

4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 2. Following the life test and while still at the maximum class temperature, electrical tests C through DD shall be performed in the order specified in 4.4.2 using the operating temperature end-of-life requirements of table I. Following the return of the module to 25°C, electrical tests C through EE shall be performed in the order specified in 4.4.2 using the 25° ±5°C ambient end-of-life requirements of table I.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-28787.

6. NOTES

6.1 Definitions. Definitions are as specified in MIL-STD-1331.

6.2 End-of-life. When any individual module circuit parameter fails to conform to the end-of-life requirements of table I, the module is considered to have completed its useful life.

* 6.3 Substitutability. Class II modules are substitutable for class I modules. When class II modules are used in a class I environment, the failure rate for class I shall be used for reliability prediction (see 3.3 and 3.4).

* 6.4 Module description. The following electrical parameters reflect the 0° to 60°C and -55° to 100°C end-of-life limits from table I.

	Class I	Class II	Unit
Quiescent power dissipation (15 V operation, outputs not loaded)	18.0 maximum	16.0 maximum	mW
Input current			
Low level (V_{IL} = 0, V_{DD} = 18 V)	-200 maximum	-200 maximum	nA
High level (V_{IH} = 15 V, V_{DD} = 18 V)	200 maximum	200 maximum	nA

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	Class I	Class II	Unit
Output voltage (V_{DD} = 5 V)			
Low level			
At -55°C (I_{OL} = 0.64 mA)	0.4 maximum	0.4 maximum	V
At 0° and 25°C (I_{OL} = 0.51 mA)	0.4 maximum	0.4 maximum	V
At 60°C (I_{OL} = 0.44 mA)	0.4 maximum	0.4 maximum	V
At 100°C (I_{OL} = 0.36 mA)	0.4 maximum	0.4 maximum	V
High level			
At -55°C (I_{OH} = 0.64 mA)	4.6 minimum	4.6 minimum	V
At 0° and 25°C (I_{OH} = 0.51 mA)	4.6 minimum	4.6 minimum	V
At 60°C (I_{OH} = 0.44 mA)	4.6 minimum	4.6 minimum	V
At 100°C (I_{OH} = 0.36 mA)	4.6 minimum	4.6 minimum	V
Output voltage (V_{DD} = 15.0 V, all outputs)			
Low level (I_{OL} = 0)	1.5 maximum	1.5 maximum	V
High level (I_{OH} = 0)	13.5 minimum	13.5 minimum	V

Input voltage (all inputs)	Classes I and II				Unit
	V_{DD} = 5.0 V		V_{DD} = 15.0 V		
Low level	Min	Max	Min	Max	
At -55°C	0.0	1.5	0.0	4.0	V
At 0° and 25°C	0.0	1.5	0.0	4.0	V
At 60°C	0.0	1.5	0.0	4.0	V
At 100°C	0.0	1.5	0.0	4.0	V
High level					
At -55°C	3.5	V_{DD}	11.0	V_{DD}	V
At 0° and 25°C	3.5	V_{DD}	11.0	V_{DD}	V
At 60°C	3.5	V_{DD}	11.0	V_{DD}	V
At 100°C	3.5	V_{DD}	11.0	V_{DD}	V

* 6.5 Module function. This module contains two independent BCD to decimal decoders and four capacitors with common grounds.

* 6.5.1 Input to output relationship. Each module shall have the input to output relationship of table II for positive logic.

6.5.2 Parallel outputs. Paralleling of outputs is not permissible.

6.5.3 Module handling considerations. To avoid possible damage to the device during handling, testing or actual operation, the following procedures should be followed:

a. Device pins should be in contact with conductive material, except when being tested or in actual operation, to avoid build-up of static charge.

b. Soldering-iron tips, metal parts of fixtures and tools, and handling facilities should be grounded.

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c. Signals should not be applied to the inputs while the device power supply is off.

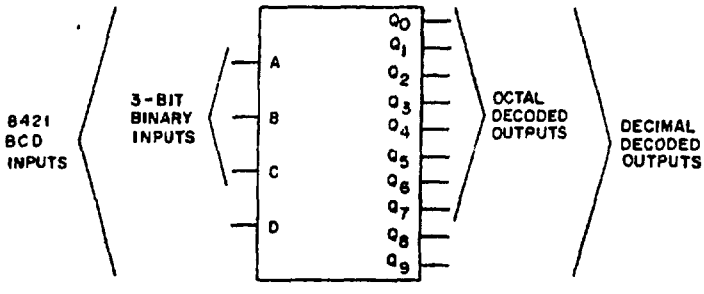
d. Input voltages greater than the supply (V_{DD} - V_{SS}) should not be applied as this will force bias the protection diodes.

e. All unused input leads must be connected to either V_{SS} (ground) or V_{DD} (device supply), whichever is appropriate for the logic circuit involved.

TABLE II. Input to output relationship. 1/

No.	Inputs				Outputs									
	D	C	B	A	Q9	Q8	Q7	Q6	Q5	Q4	Q3	Q2	Q1	Q0
0	L	L	L	L	L	L	L	L	L	L	L	L	L	H
1	L	L	L	H	L	L	L	L	L	L	L	L	H	L
2	L	L	H	L	L	L	L	L	L	L	L	H	L	L
3	L	L	H	H	L	L	L	L	L	L	H	L	L	L
4	L	H	L	L	L	L	L	L	L	H	L	L	L	L
5	L	H	L	H	L	L	L	L	H	L	L	L	L	L
6	L	H	H	L	L	L	L	H	L	L	L	L	L	L
7	L	H	H	H	L	L	H	L	L	L	L	L	L	L
8	H	L	L	L	L	H	L	L	L	L	L	L	L	L
9	H	L	L	H	H	L	L	L	L	L	L	L	L	L

1/ H = High-voltage level.
L = Low-voltage level.



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6.5.4 Dynamic power dissipation. The dynamic power dissipation can be approximated by the formula:

$$P_{Total} = (CV_{DD}^2F)N + \text{quiescent power dissipation.}$$

Where: C = Load capacitance.
V_{DD} = Supply voltage.
F = Operating frequency.
N = Number of outputs.

6.6 Changes from previous issue. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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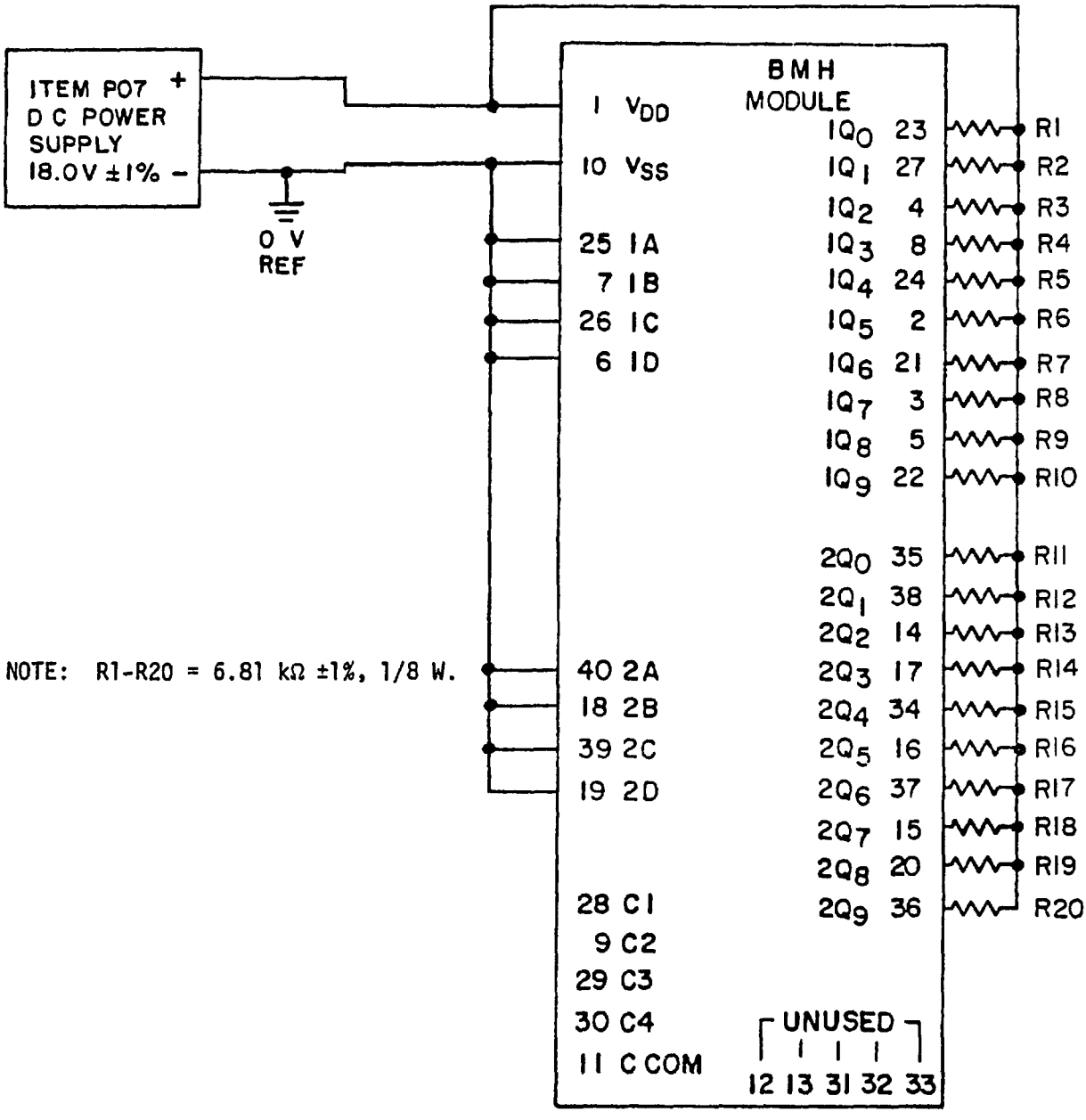
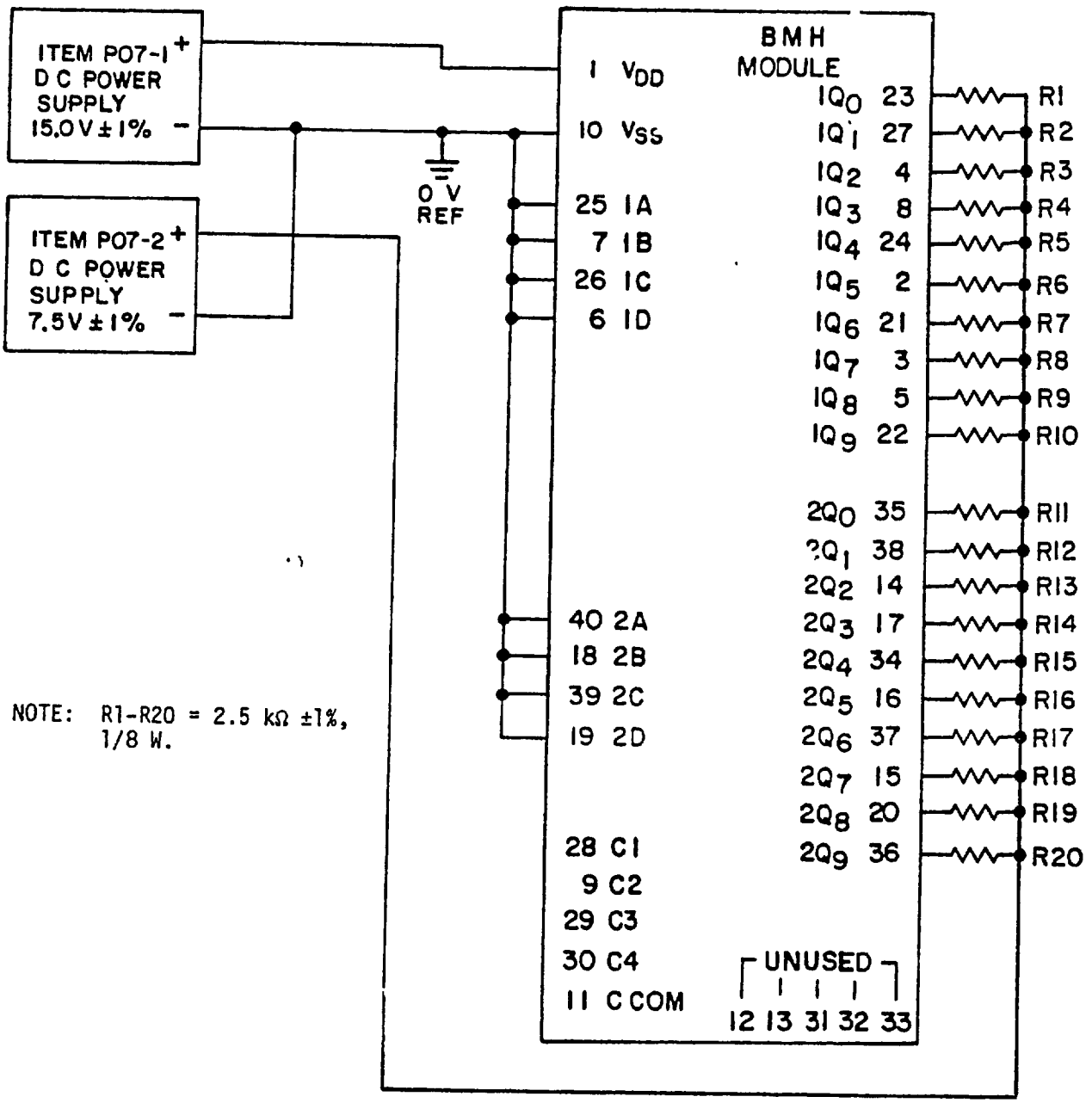


FIGURE 2. Operating temperature and life test circuit.

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NOTE: R1-R20 = 2.5 kΩ ±1%, 1/8 W.

FIGURE 3. Component temperature test circuit.

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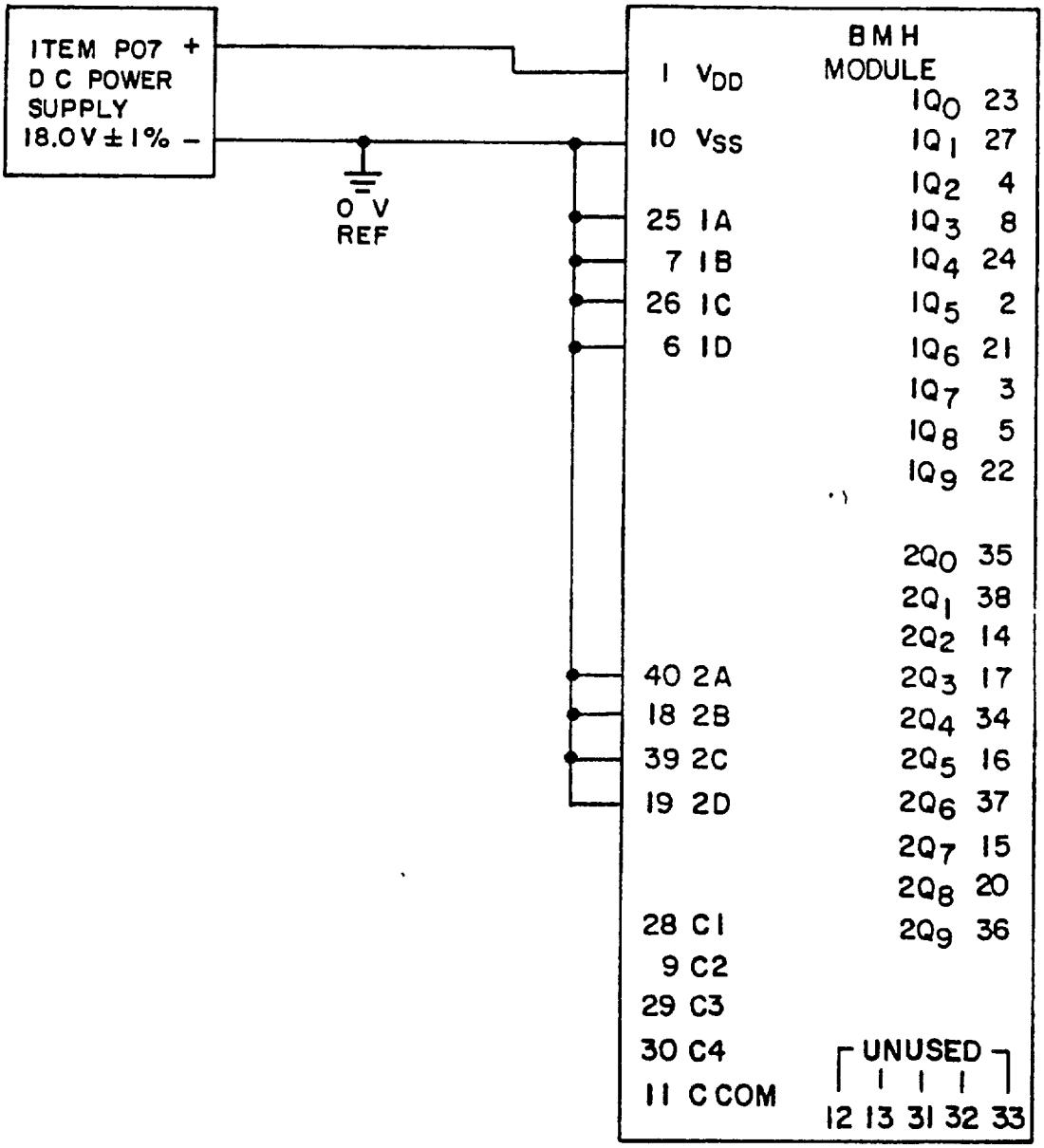


FIGURE 4. Vibration test circuit.

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TABLE III. Isolation test.

Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
A1	Isolation	C,12,13,31,32,33	Module frame	10	∞	MΩ	5
A2	↓	12	C	↓	↓	↓	↓
A3		13	↓				
A4		31					
A5		32					
A6		33					

TABLE IV. Negative input clamp diode voltage test.

Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
C1	Negative input clamp diode voltage	25	Open	-0.4	-6.0	V	6
C2	↓	7	↓	↓	↓	↓	↓
C3		26					
C4		6					
C5		40					
C6		18					
C7		39					
C8		19					

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TABLE V. Positive input clamp diode voltage test.

Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
D1	Positive input clamp diode voltage	Open	25	0.4	1.5	V	6
D2	↓	↓	7	↓	↓	↓	↓
D3			26				
D4			6				
D5			40				
D6			18				
D7			39				
D8			19				

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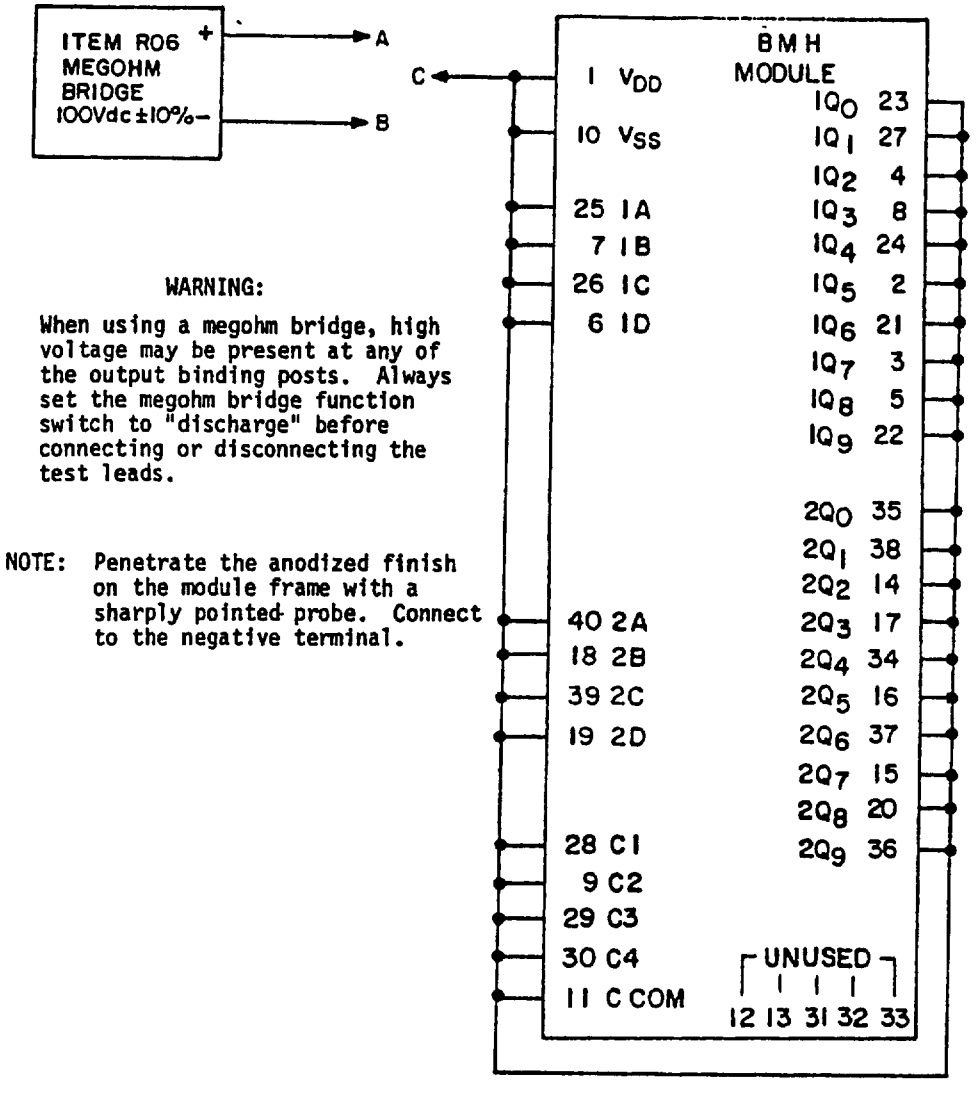
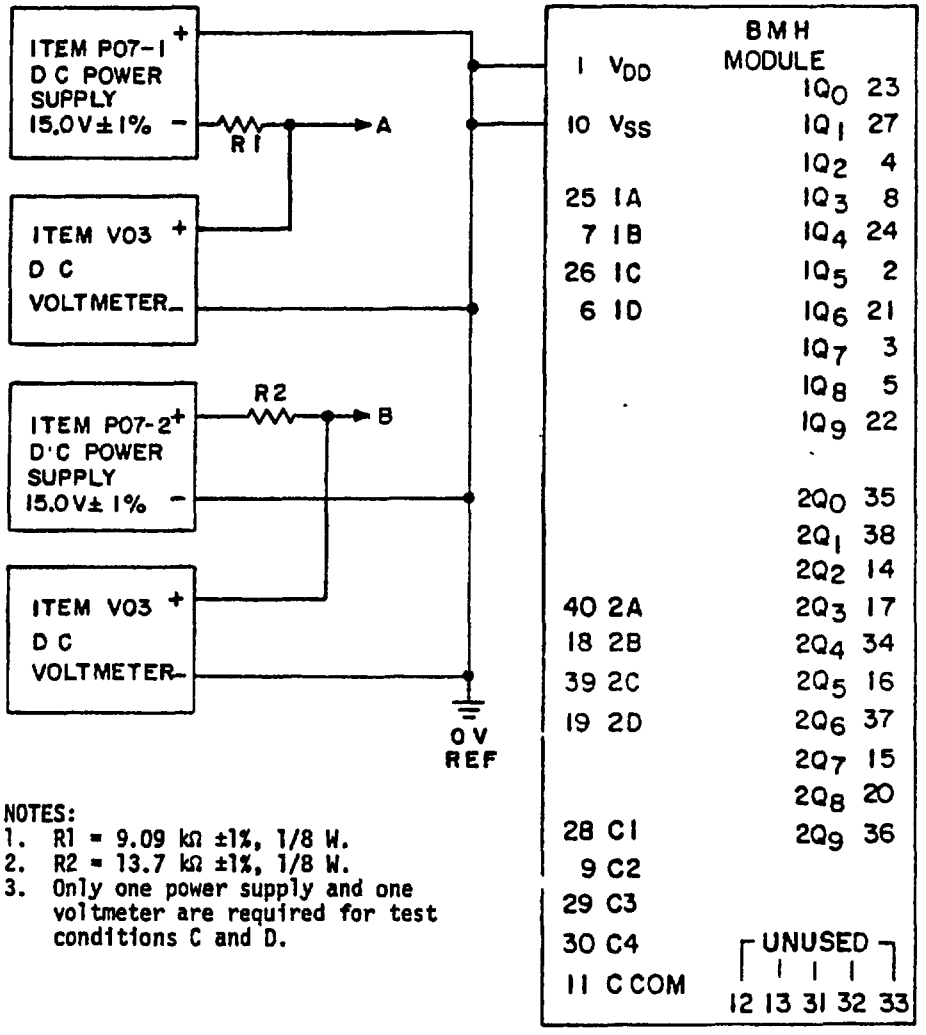


FIGURE 5. Isolation test circuit for table III.

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NOTES:
1. R1 = 9.09 kΩ ±1%, 1/8 W.
2. R2 = 13.7 kΩ ±1%, 1/8 W.
3. Only one power supply and one voltmeter are required for test conditions C and D.

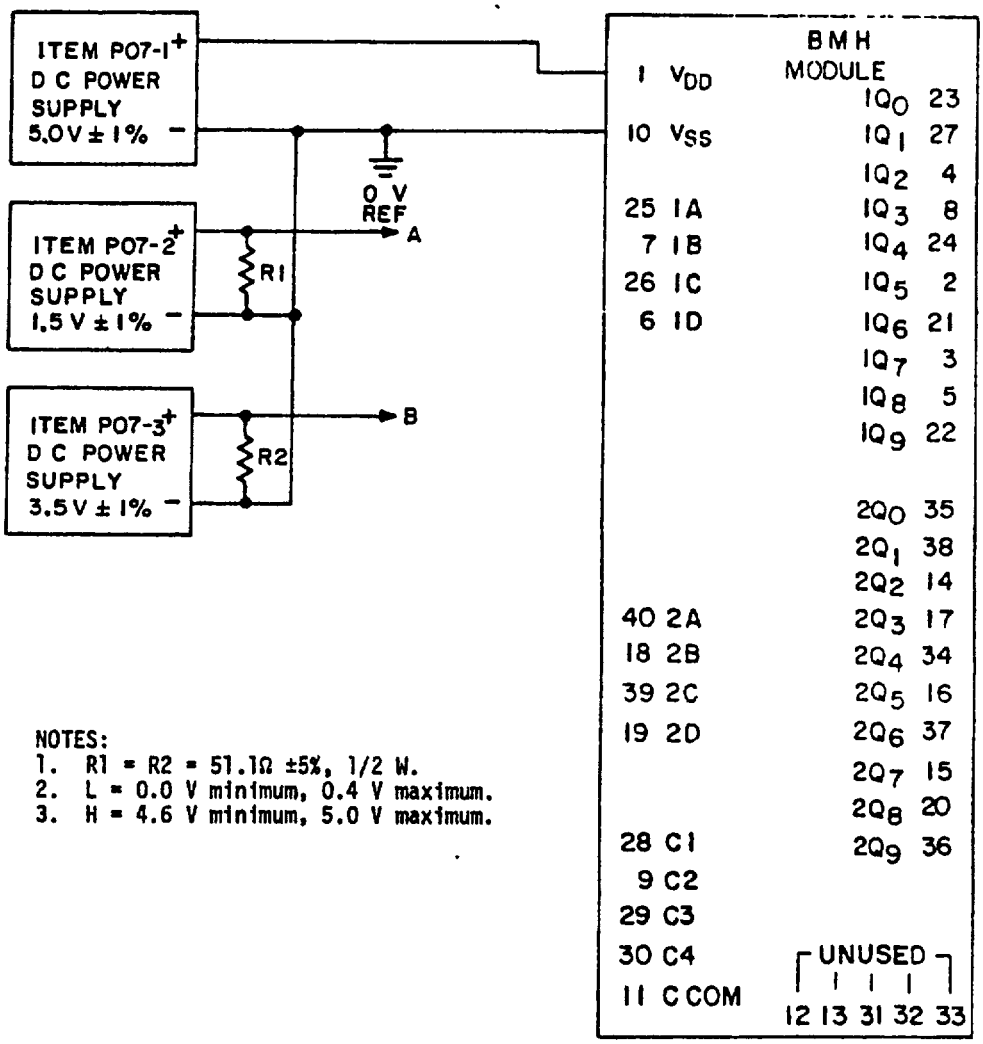
FIGURE 6. Positive and negative input clamp diode voltage test circuit for tables IV and V.

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TABLE VI. Functional test.

Test condition	Parameter	Connect		Output pins																				Test figure
		A to pin	B to pin	23	27	4	8	24	2	21	3	5	22	35	38	14	17	34	16	37	15	20	36	
E1	Functional	25,7,26,6,40,18, 39,19	Open	H	L	L	L	L	L	L	L	L	L	H	L	L	L	L	L	L	L	L	L	7
E2		7,26,6,18,39,19	25,40	L	H	L	L	L	L	L	L	L	L	L	H	L	L	L	L	L	L	L	L	
E3		25,26,6,40,39,19	7,18	L	L	H	L	L	L	L	L	L	L	L	L	H	L	L	L	L	L	L	L	
E4		26,6,39,19	25,7,40,18	L	L	L	H	L	L	L	L	L	L	L	L	L	H	L	L	L	L	L	L	
E5		25,7,6,40,18,19	26,39	L	L	L	L	H	L	L	L	L	L	L	L	L	L	H	L	L	L	L	L	
E6		7,6,18,19	25,26,40,39	L	L	L	L	L	H	L	L	L	L	L	L	L	L	L	H	L	L	L	L	
E7		25,6,40,19	7,26,18,39	L	L	L	L	L	L	H	L	L	L	L	L	L	L	L	L	H	L	L	L	
E8		6,19	25,7,26,40, 18,39	L	L	L	L	L	L	L	H	L	L	L	L	L	L	L	L	L	H	L	L	
E9		25,7,26,40,18,39	6,19	L	L	L	L	L	L	L	L	H	L	L	L	L	L	L	L	L	L	H	L	
E10		7,26,18,39	25,6,40,19	L	L	L	L	L	L	L	L	L	H	L	L	L	L	L	L	L	L	L	H	

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NOTES:
1. R1 = R2 = 51.1Ω ±5%, 1/2 W.
2. L = 0.0 V minimum, 0.4 V maximum.
3. H = 4.6 V minimum, 5.0 V maximum.

FIGURE 7. Functional test circuit for table VI.

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TABLE VII. Static output voltage (low-level) test.

Test condition	Parameter		Connect			Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	Min	Max		
F1	1Q0	Static output voltage, low level (I_{OL} = 0)	23	7,26,6,40, 18,39,19	25	0.0	0.05	V	8
F2	1Q1		27	25,7,26,6, 40,18,39,19	Open				
F3	1Q2		4						
F4	1Q3		8						
F5	1Q4		24						
F6	1Q5		2						
F7	1Q6		21						
F8	1Q7		3						
F9	1Q8		5						
F10	1Q9		22						
F11	2Q0		35	25,7,26,6, 18,39,19	40				
F12	2Q1		38	25,7,26,6, 40,18,39,19	Open				
F13	2Q2		14						
F14	2Q3		17						
F15	2Q4		34						
F16	2Q5		16						
F17	2Q6		37						
F18	2Q7		15						
F19	2Q8		20						
F20	2Q9		36						

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TABLE VIII. Static output voltage (high-level) test.

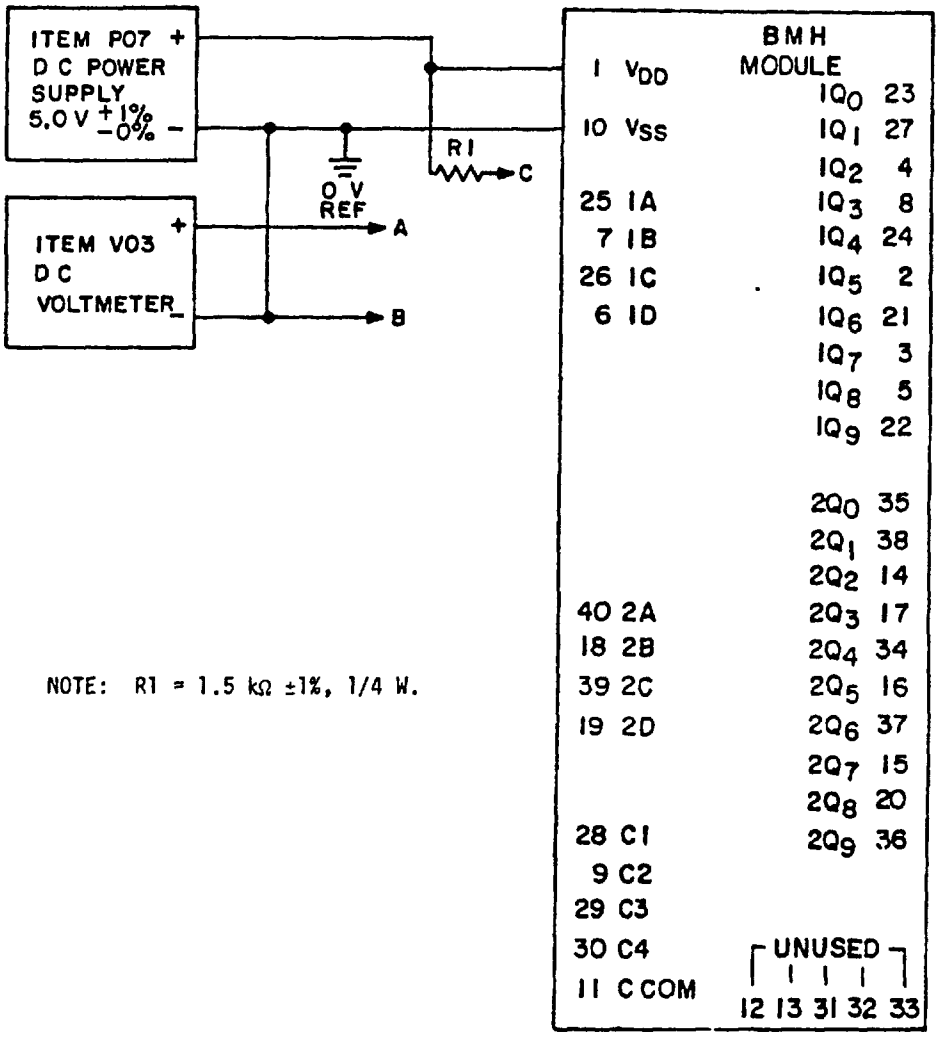
Test condition	Parameter		Connect			Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	Min	Max		
G1	1Q0	Static output voltage, high level (I_{OH} = 0)	23	25,7,26,6, 40,18,39,19	Open	4.95	5.05	V	8
G2	1Q1		27	7,26,6,40, 18,39,19	25				
G3	1Q2		4	25,26,6,40, 18,39,19	7				
G4	1Q3		8	26,6,40,18, 39,19	7,25				
G5	1Q4		24	25,7,6,40, 18,39,19	26				
G6	1Q5		2	7,6,40,18, 39,19	25,26				
G7	1Q6		21	25,6,40,18, 39,19	26,7				
G8	1Q7		3	6,40,18,39, 19	25,26,7				
G9	1Q8		5	25,7,26,40, 18,39,19	6				
G10	1Q9		22	7,26,40,18, 39,19	25,6				
G11	2QC		35	25,7,26,6, 40,18,39,19	Open				
G12	2Q1		38	25,7,26,6, 18,39,19	40				

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TABLE VIII. Static output voltage (high-level) test - Continued.

Test condition	Parameter		Connect			Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	Min	Max		
G13	2Q2	Static output voltage, high level (I_{OH} = 0)	14	25,7,26,6, 40,39,19	18	4.95	5.05	V	8
G14	2Q3		17	25,7,26,6, 39,19	40,18				
G15	2Q4		34	25,7,26,6, 40,18,19	39				
G16	2Q5		16	25,7,26,6, 18,19	39,40				
G17	2Q6		37	25,7,26,6, 40,19	18,39				
G18	2Q7		15	25,7,26,6, 19	18,39,40				
G19	2Q8		20	25,7,26,6, 40,18,39	19				
G20	2Q9		36	25,7,26,6, 18,39	19,40				

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NOTE: R1 = 1.5 kΩ ±1%, 1/4 W.

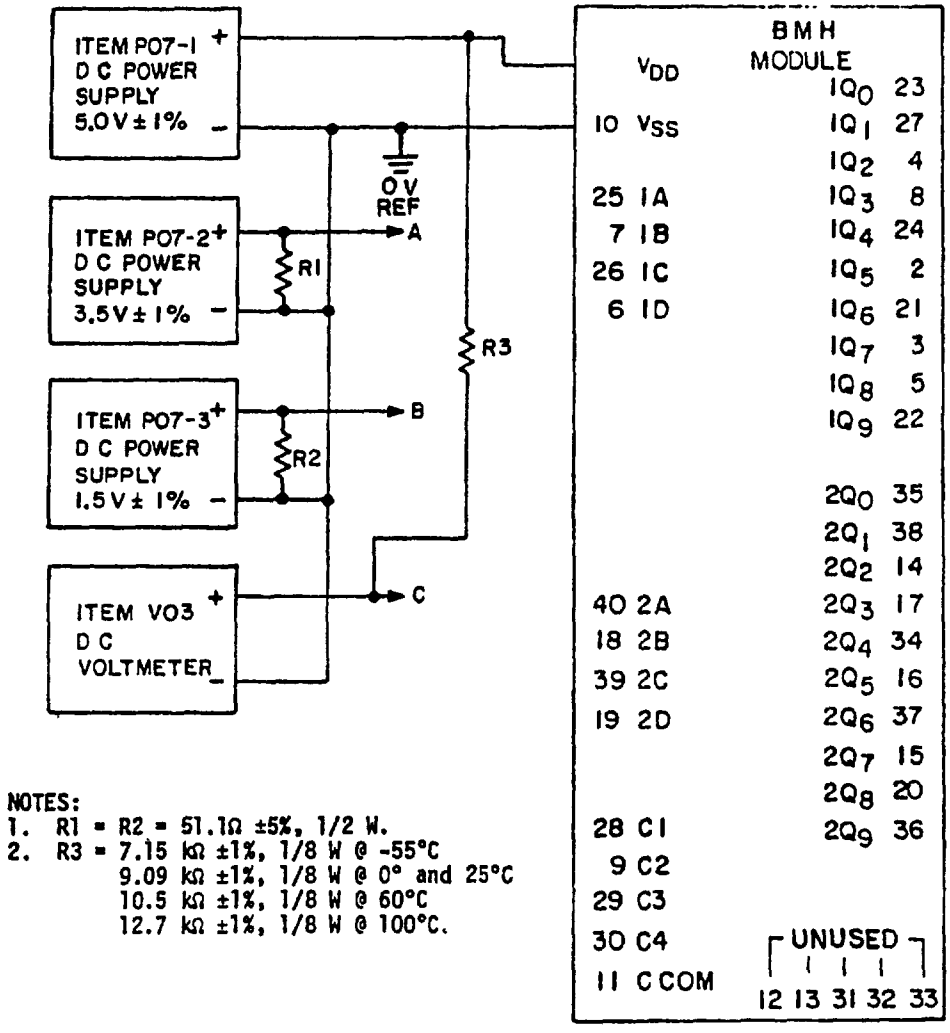
*FIGURE 8. Static output voltage (low-level and high-level) test circuit for tables VII and VIII.

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TABLE IX. Static output voltage (low-level) test.

Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
H1	Static output voltage, low level (V_{DD} = 5 V)	25,40	7,26,6,18, 39,19	23	0.0	0.4	V	9
H2		Open	25,7,26,6, 40,18,39,19	27				
H3				4				
H4				8				
H5				24				
H6				2				
H7				21				
H8				3				
H9				5				
H10				22				
H11		25,40	7,26,6,18, 39,19	35				
H12		Open	25,7,26,6, 40,18,39,19	38				
H13				14				
H14				17				
H15				34				
H16				16				
H17				37				
H18				15				
H19				20				
H20				36				

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NOTES:
1. R1 = R2 = 51.1Ω ±5%, 1/2 W.
2. R3 = 7.15 kΩ ±1%, 1/8 W @ -55°C
   9.09 kΩ ±1%, 1/8 W @ 0° and 25°C
   10.5 kΩ ±1%, 1/8 W @ 60°C
   12.7 kΩ ±1%, 1/8 W @ 100°C.

FIGURE 9. Static output voltage (low-level) test circuit for table IX.

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TABLE X. Static output voltage (high-level) test.

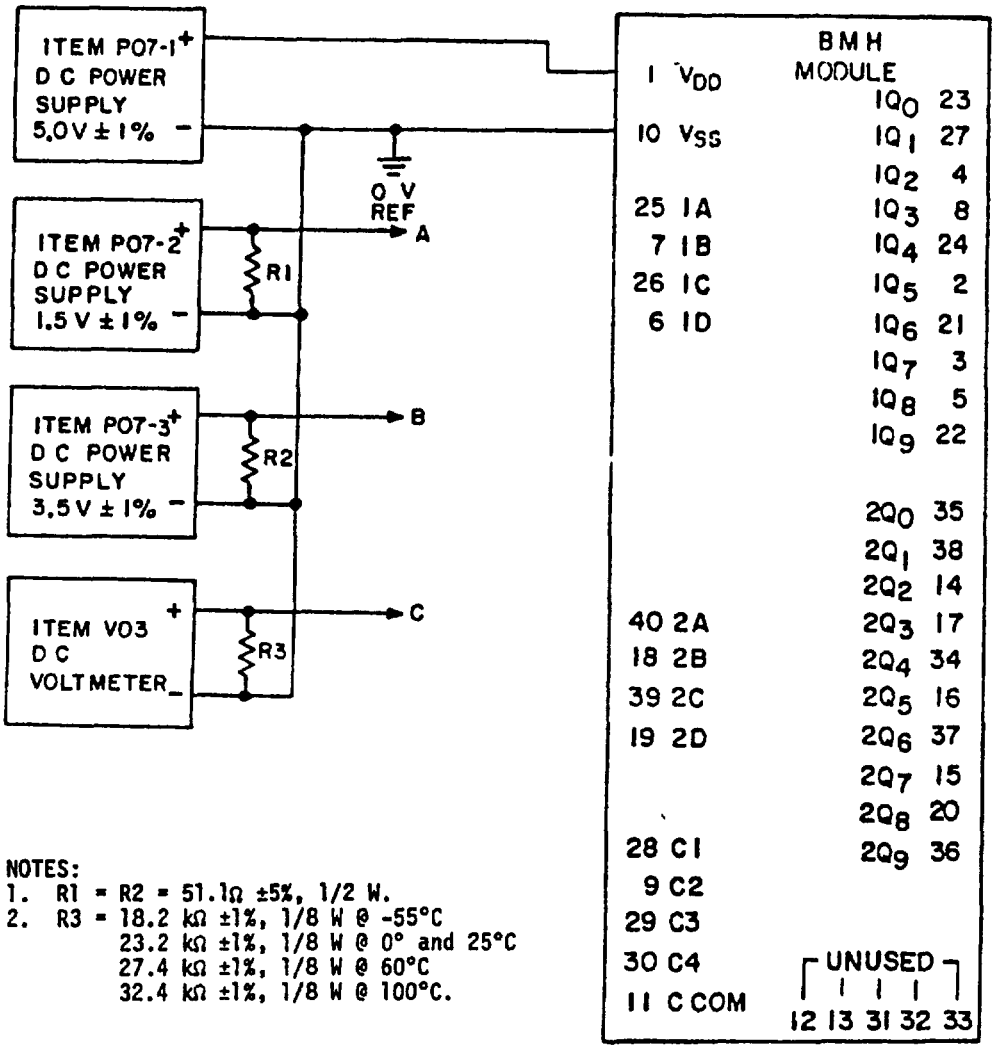
Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
J1	Static output voltage, high level (V_{DD} = 5 V)	25,7,26,6, 40,18,39,19	Open	23	4.6	5.0	V	10
J2		7,26,6,18, 39,19	25,40	27				
J3		25,26,6,40, 39,19	7,18	4				
J4		26,6,39,19	25,7,40,18	8				
J5		25,7,6,40, 18,19	26,39	24				
J6		7,6,18,19	25,26,40,39	2				
J7		25,6,40,19	7,26,18,39	21				
J8		6,19	25,7,26,40, 18,39	3				
J9		25,7,26,40, 18,39	6,19	5				
J10		7,26,18,39	25,6,40,19	22				
J11		25,7,26,6, 40,18,39,19	Open	35				
J12		7,26,6,18, 39,19	25,40	38				
J13		25,26,6,40, 39,19	7,18	14				
J14		26,6,39,19	25,7,40,18	17				

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TABLE X. Static output voltage (high-level) test - Continued.

Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
J15	Static output voltage, high level (V_{DD} = 5 V)	25,7,6,40, 18,19	26,39	34	4.6	5.0	V	10
J16		7,6,18,19	25,26,40,39	16				
J17		25,6,40,19	7,26,18,39	37				
J18		6,19	25,7,26,40, 18,39	15				
J19		25,7,26,40, 18,39	6,19	20				
J20		7,26,18,39	25,6,40,19	36				

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NOTES:
1. R1 = R2 = 51.1Ω ±5%, 1/2 W.
2. R3 = 18.2 kΩ ±1%, 1/8 W @ -55°C
   23.2 kΩ ±1%, 1/8 W @ 0° and 25°C
   27.4 kΩ ±1%, 1/8 W @ 60°C
   32.4 kΩ ±1%, 1/8 W @ 100°C.

FIGURE 10. Static output voltage (high-level) test circuit for table X.

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TABLE XI. Static output voltage (low-level) test.

Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
K1	Static output voltage, low level (V_{DD} = 15.0 V)	7,26,6,18, 39,19	25,40	23	0.0	1.5	V	11
K2		25,7,26,6, 40,18,39,19	Open	27				
K3				4				
K4				8				
K5				24				
K6				2				
K7				21				
K8				3				
K9				5				
K10				22				
K11		7,26,6,18, 39,19	25,40	35				
K12		25,7,26,6, 40,18,39,19	Open	38				
K13				14				
K14				17				
K15				34				
K16				16				
K17				37				
K18				15				
K19				20				
K20				36				

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TABLE XII. Static output voltage (high-level) test.

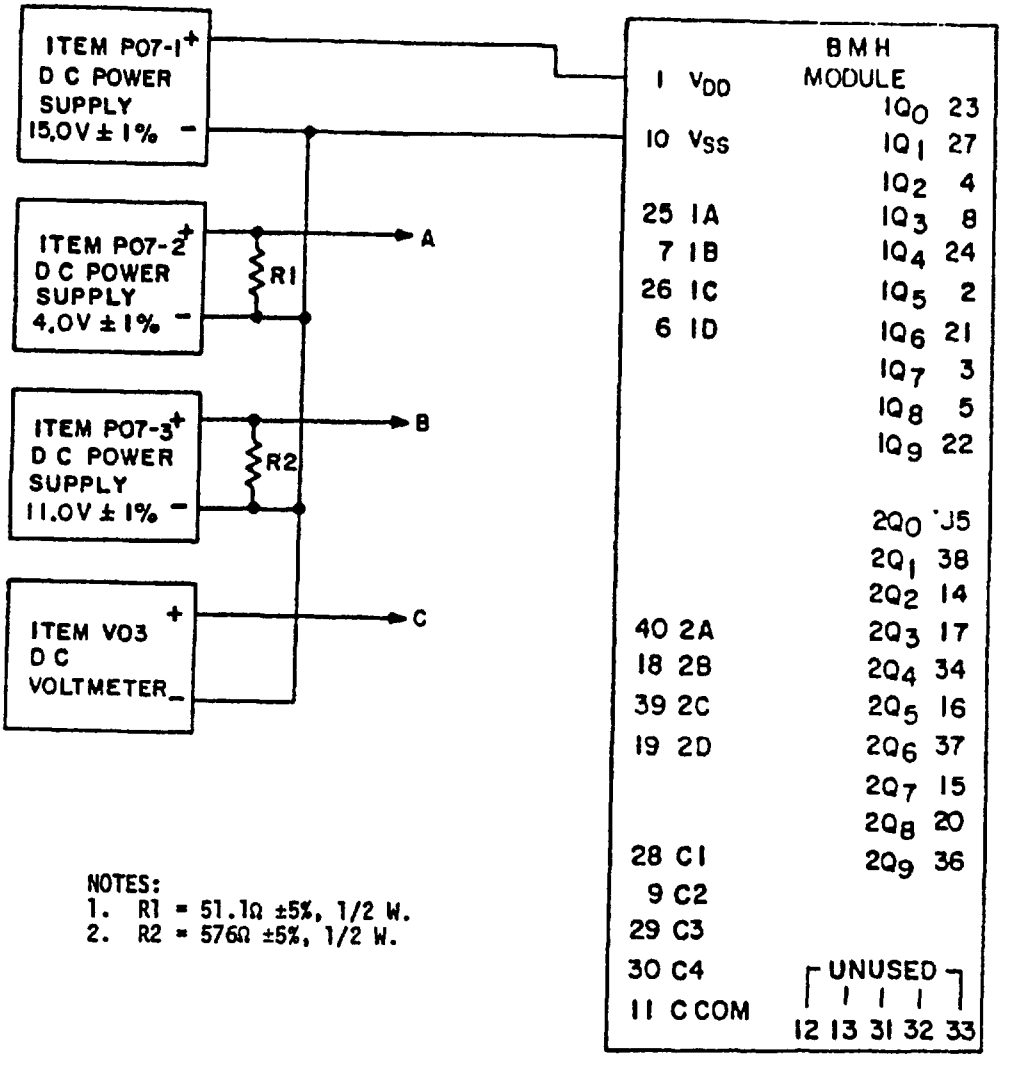
Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
L1	Static output voltage, high level (V_{DD} = 15.0 V)	25,7,26,6, 40,18,39,19	Open	23	13.5	15.0	V	11
L2		7,26,6,18, 39,19	25,40	27				
L3		25,26,6,40, 39,19	7,18	4				
L4		26,6,39,19	25,7,40,18	8				
L5		25,7,6,40, 18,19	26,39	24				
L6		7,6,18,19	25,26,40,39	2				
L7		25,6,40,19	7,26,18,39	21				
L8		6,19	25,7,26,40, 18,39	3				
L9		25,7,26,40, 18,39	6,19	5				
L10		7,26,18,39	25,6,40,19	22				
L11		25,7,26,6, 40,18,39,19	Open	35				
L12		7,26,6,18, 39,19	25,40	38				
L13		25,26,6,40, 39,19	7,18	14				
L14		26,6,39,19	25,7,40,18	17				

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TABLE XII. Static output voltage (high-level) test - Continued.

Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
L15	Static output voltage, high level (V_{DD} = 15.0 V)	25,7,6,40, 18,19	26,39	34	13.5	15.0	V	11
L16	↓	7,6,18,19	25,26,40,39	16	↓	↓	↓	↓
L17		25,6,40,19	7,26,18,39	37				
L18		6,19	25,7,26,40, 18,39	15				
L19		25,7,26,40, 18,39	6,19	20				
L20		7,26,18,39	25,6,40,19	36				

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NOTES:
1. R1 = 51.1Ω ±5%, 1/2 W.
2. R2 = 576Ω ±5%, 1/2 W.

FIGURE 11. Static output voltage (low-level and high-level) test circuit for tables XI and XII.

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TABLE XIII. Input current (high-level) test.

Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
M1	Input current, high level	25	7,26,5,40,18, 39,19	0	200	nA	12
M2	↓	7	25,26,6,40,18, 39,19	↓	↓	↓	↓
M3		26	25,7,6,40,18, 39,19				
M4		6	25,7,26,40,18, 39,19				
M5		40	25,7,26,6,18, 39,19				
M6		18	25,7,26,6,40, 39,19				
M7		39	25,7,26,6,40, 18,19				
M8		19	25,7,26,6,40, 18,39				

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TABLE XIV. Input current (low-level) test.

Test condition	Parameter	Connect A to pin	Connect B to pin	Initial limits 25°±5°C ambient Min	Initial limits 25°±5°C ambient Max	Unit	Test figure
N1	Input current, low level	25	7,26,6,40,18, 39,19	0	-200	nA	13
N2	↓	7	25,26,6,40,18, 39,19	↓	↓	↓	↓
N3		26	25,7,6,40,18, 39,19				
N4		6	25,7,26,40,18, 39,19				
N5		40	25,7,26,6,18, 39,19				
N6		18	25,7,26,6,40, 39,19				
N7		39	25,7,26,6,40, 18,19				
N8		19	25,7,26,6,40, 18,39				

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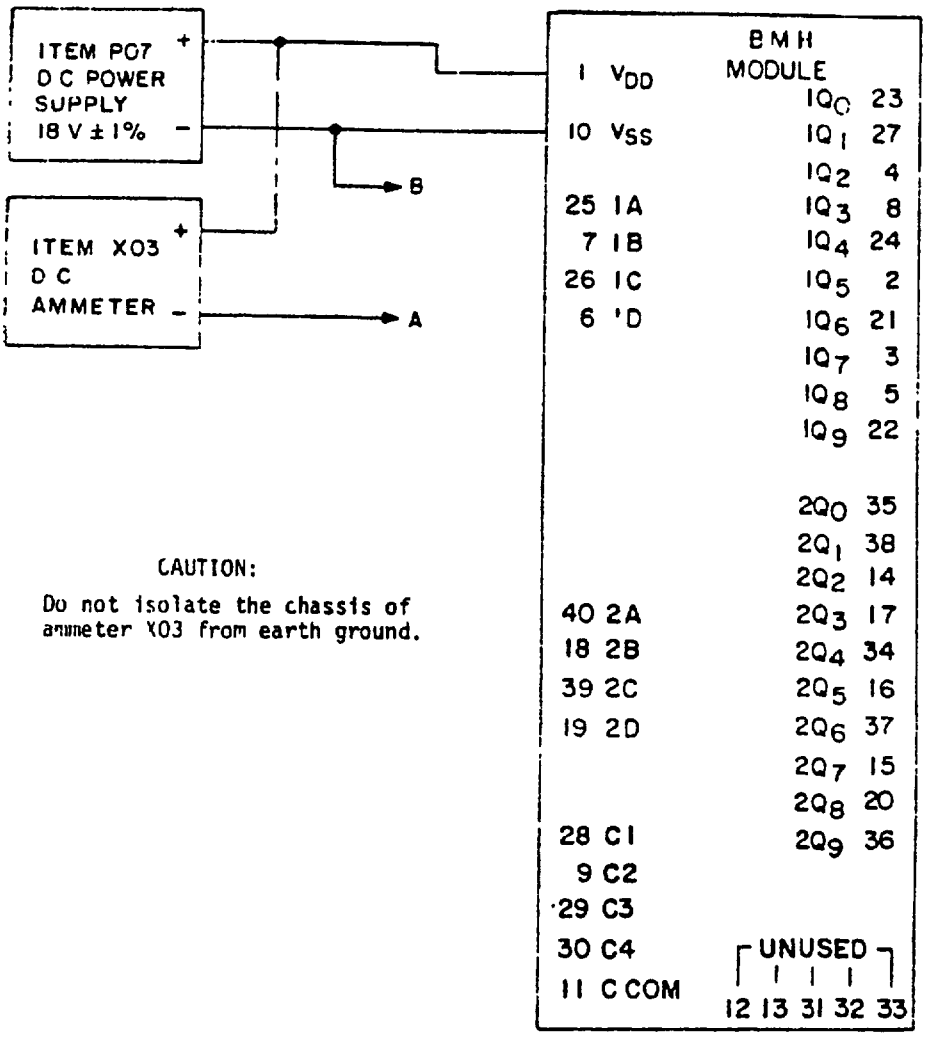


FIGURE 12. Input current (high-level) test circuit for table XIII.

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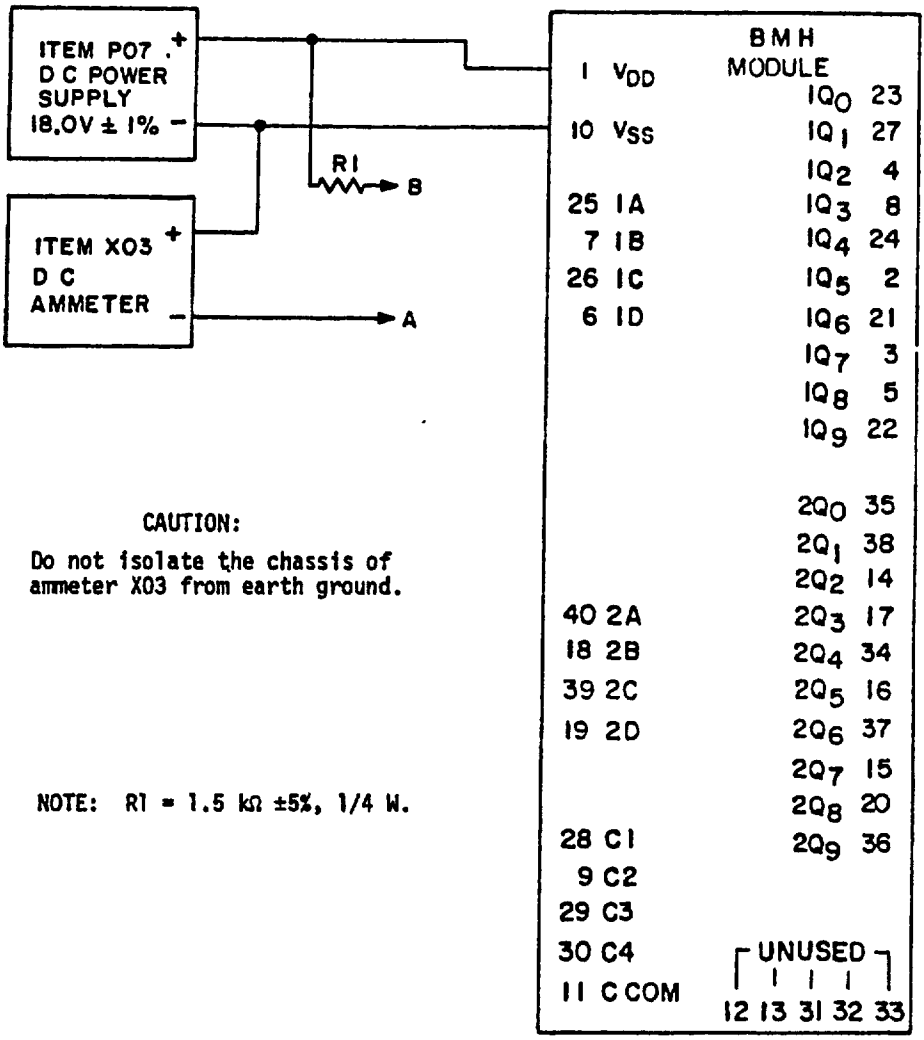


FIGURE 13. Input current (low-level) test circuit for table XIV.

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TABLE XV. Static power supply current (low-level and high-level) test.

Test condition	Parameter	Initial limits 25°±5°C ambient		Unit	Test figure
		Min	Max		
P	Static power supply current	0	40	µA	14

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TABLE XVI. Transition time (t_{TLH}) test.

Test condition	Parameter	Connect				Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
Q1	Transition time (t_{TLH}), C_L = 50 pF, V_{DD} = 5 V	Open	7,26,6,40, 18,39,19	25	23	35	350	ns	15,17
Q2		Open	7,26,6,40, 18,39,19	25	27				15,16
Q3		Open	25,26,6,40, 18,39,19	7	4				
Q4		25	26,6,40,18, 39,19	7	8				
Q5		Open	25,7,6,40, 18,39,19	26	24				
Q6		25	7,6,40,18, 39,19	26	2				
Q7		7	25,6,40,18, 39,19	26	21				
Q8		25,7	6,40,18,39, 19	26	3				
Q9		Open	25,7,26,40, 18,39,19	6	5				
Q10		25	7,26,40,18, 39,19	6	22				
Q11		Open	25,7,26,6, 18,39,19	40	35				15,17
Q12		Open	25,7,26,6, 18,39,19	40	38				15,16
Q13		Open	25,7,26,6, 40,39,19	18	14				15,16

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TABLE XVI. Transition time (t_{TLH}) test - Continued.

Test condition	Parameter	Connect				Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
Q14	Transition time (t_{TLH}), C_L = 50 pF, V_{DD} = 5 V	40	25,7,26,6, 39,19	18	17	35	350	ns	15,16
Q15	↓	Open	25,7,26,6, 40,18,19	39	34	↓	↓	↓	↓
Q16		40	25,7,26,6, 18,19	39	16				
Q17		18	25,7,26,6, 40,19	39	37				
Q18		40,18	25,7,26,6,19	39	15				
Q19		Open	25,7,26,6, 40,18,39	19	20				
Q20		40	25,7,26,6, 18,39	19	36				
Q21	Transition time (t_{TLH}), C_L = 150 pF, V_{DD} = 5 V	Open	7,26,6,40, 18,39,19	25	23	83	830		15,17
Q22	↓	Open	7,26,6,40, 18,39,19	25	27	↓	↓	↓	15,16
Q23		Open	25,26,6,40, 18,39,19	7	4				
Q24		25	26,6,40,18, 39,19	7	8				
Q25		Open	25,7,6,40, 18,39,19	26	24				
Q26		25	7,6,40,18, 39,19	26	2				

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TABLE XVI. Transition time (t_{TLH}) test - Continued.

Test condition	Parameter	Connect				Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
Q27	Transition time (t_{TLH}), C_L = 150 pF, V_{DD} = 5 V	7	25,6,40,18, 39,19	26	21	83	830	ns	15,16
Q28		25,7	6,40,18,39, 19	26	3				
Q29		Open	25,7,26,40, 18,39,19	6	5				
Q30		25	7,26,40,18, 39,19	6	22				
Q31		Open	25,7,26,6, 18,39,19	40	35				15,17
Q32		Open	25,7,26,6, 18,39,19	40	38				15,16
Q33		Open	25,7,26,6, 40,39,19	18	14				
Q34		40	25,7,26,6, 39,19	18	17				
Q35		Open	25,7,26,6, 40,18,19	39	34				
Q36		40	25,7,26,6, 18,19	39	16				
Q37		18	25,7,26,6, 40,19	39	37				
Q38		40,18	25,7,26,6,19	39	15				
Q39		Open	25,7,26,6, 40,18,39	19	20				
Q40		40	25,7,26,6, 18,39	19	36				

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TABLE XVII. Transition time (t_{THL}) test.

Test condition	Parameter	Connect				Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
R1	Transition time (t_{THL}), C_L = 50 pF, V_{DD} = 5 V	Open	7,26,6,40, 18,39,19	25	23	17	175	ns	15,17
R2		Open	7,26,6,40, 18,39,19	25	27				15,16
R3		Open	25,26,6,40, 18,39,19	7	4				
R4		25	26,6,40,18, 39,19	7	8				
R5		Open	25,7,6,40, 18,39,19	26	24				
R6		25	7,6,40,18, 39,19	26	2				
R7		7	25,6,40,18, 39,19	26	21				
R8		25,7	6,40,18,39, 19	26	3				
R9		Open	25,7,26,40, 18,39,19	6	5				
R10		25	7,26,40,18, 39,19	6	22				
R11		Open	25,7,26,6, 18,39,19	40	35				15,17
R12		Open	25,7,26,6, 18,39,19	40	38				15,16
R13		Open	25,7,26,6, 40,39,19	18	14				15,16

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TABLE XVII. Transition time (t_{THL}) test - Continued.

Test condition	Parameter	Connect				Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
R14	Transition time (t_{THL}), C_L = 50 pF, V_{DD} = 5 V	40	25,7,26,6, 39,19	18	17	17	175	ns	15,16
R15	↓	Open	25,7,26,6, 40,18,19	39	34	↓	↓	↓	↓
R16		40	25,7,26,6, 18,19	39	16				
R17		18	25,7,26,6, 40,19	39	37				
R18		40,18	25,7,26,6,19	39	15				
R19		Open	25,7,26,6, 40,18,39	19	20				
R20		40	25,7,26,6, 18,39	19	36				
R21	Transition time (t_{THL}), C_L = 150 pF, V_{DD} = 5 V	Open	7,26,6,40, 18,39,19	25	23	65	655		15,17
R22		Open	7,26,6,40, 18,39,19	25	27				15,16
R23		Open	25,26,6,40, 18,39,19	7	4				
R24		25	26,6,40,18, 39,19	7	8				
R25		Open	25,7,6,40, 18,39,19	26	24				
R26	↓	25	7,6,40,18, 39,19	26	2	↓	↓	↓	↓

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TABLE XVII. Transition time (t_{THL}) test - Continued.

Test condition	Parameter	Connect				Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
R27	Transition time (t_{THL}), C_L = 150 pF, V_{DD} = 5 V	7	25,6,40,18,39,19	26	21	65	655	ns	15,16
R28		25,7	6,40,18,39,19	26	3				
R29		Open	25,7,26,40,18,39,19	6	5				
R30		25	7,26,40,18,39,19	6	22				
R31		Open	25,7,26,6,18,39,19	40	35				15,17
R32		Open	25,7,26,6,18,39,19	40	38				15,16
R33		Open	25,7,26,6,40,39,19	18	14				
R34		40	25,7,26,6,39,19	18	17				
R35		Open	25,7,26,6,40,18,19	39	34				
R36		40	25,7,26,6,18,19	39	16				
R37		18	25,7,26,6,40,19	39	37				
R38		40,18	25,7,26,6,19	39	15				
R39		Open	25,7,26,6,40,18,39	19	20				
R40		40	25,7,26,6,18,39	19	36				

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* TABLE XVIII. Propagation delay time (t_{PLH}) test.

Test condition	Parameter	Connect				Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
S1	Propagation delay time (t_{PLH}), V_{DD} = 5 V, C_L = 50 pF	Open	7,26,6,40, 18,39,19	25	23	48	480	ns	15,17
S2		Open	7,26,6,40, 18,39,19		27				15,16
S3		7	26,6,40,18, 39,19		4				15,17
S4		7	26,6,40,18, 39,19		8				15,16
S5		26	7,6,40,18, 39,19		24				15,17
S6		26	7,6,40,18, 39,19		2				15,16
S7		7,26	6,40,18,39, 19		21				15,17
S8		7,26	6,40,18,39, 19		3				15,16
S9		6	7,26,40,18, 39,19		5				15,17
S10		6	7,26,40,18, 39,19		22				15,16
S11		Open	25,26,6,40, 18,39,19	7	23				15,17
S12		25	26,6,40,18, 39,19		27				15,17
S13		Open	25,26,6,40, 18,39,19		4				15,16

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* TABLE XVIII. Propagation delay time (t_{PLH}) test - Continued.

Test condition	Parameter	Connect				Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
S14	Propagation delay time (t_{PLH}), V_{DD} = 5 V, C_L = 50 pF	25	26,6,40,18, 39,19	7	8	48	480	ns	15,16
S15		26	25,6,40,18, 39,19		24				15,17
S16		25,26	6,40,18,39, 19		2				15,17
S17		26	25,6,40,18, 39,19		21				15,16
S18		25,26	6,40,18,39, 19		3				15,16
S19		Open	25,7,6,40, 18,39,19	26	23				15,17
S20		25	7,6,40,18, 39,19		27				
S21		7	25,6,40,18, 39,19		4				
S22		7,25	6,40,18,39, 19		8				
S23		Open	25,7,6,40, 18,39,19		24				15,16
S24		25	7,6,40,18, 39,19		2				
S25		7	25,6,40,18, 39,19		21				
S26		7,25	6,40,18,39, 19		3				

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* TABLE XVIII. Propagation delay time (t_{PLH}) test - Continued.

Test condition	Parameter	Connect				Initial limits 25°±5°C ambient		Unit	Test. figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
S27	Propagation delay time (t_{PLH}), V_{DD} = 5 V, C_L = 50 pF	Open	25,7,26,40, 18,39,19	6	23	48	480	ns	15,17
S28	↓	25	7,26,40,18, 39,19	↓	27	↓	↓	↓	15,17
S29		Open	25,7,26,40, 18,39,19		5				15,16
S30		25	7,26,40,18, 39,19		22				15,16
S31		Open	25,7,26,6, 18,39,19	40	35				15,17
S32		Open	25,7,26,6, 18,39,19		38				15,16
S33		18	25,7,26,6, 39,19		14				15,17
S34		18	25,7,26,6, 39,19		17				15,16
S35		39	25,7,26,6, 18,19		34				15,17
S36		39	25,7,26,6, 18,19		16				15,16
S37		18,39	27,7,26,6,19		37				15,17
S38		18,39	25,7,26,6,19		15				15,16
S39		19	25,7,26,6, 18,39		20				15,17
S40		19	25,7,26,6, 18,39		36				15,16

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* TABLE XVIII. Propagation delay time (t_{PLH}) test - Continued.

Test condition	Parameter	Connect				Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
S41	Propagation delay time (t_{PLH}), V_{DD} = 5 V, C_L = 50 pF	Open	25,7,26,6, 40,39,19	18	35	48	480	ns	15,17
S42		40	25,7,26,6, 39,19		38				15,17
S43		Open	25,7,26,6, 40,39,19		14				15,16
S44		40	25,7,26,6, 39,19		17				15,16
S45		39	25,7,26,6, 40,19		34				15,17
S46		39,40	25,7,26,6,19		16				15,17
S47		39	25,7,26,6, 40,19		37				15,16
S48		39,40	25,7,26,6,19		15				15,16
S49		Open	25,7,26,6, 40,18,19	39	35				15,17
S50		40	25,7,26,6, 18,19		38				
S51		18	25,7,26,6, 40,19		14				
S52		18,40	25,7,26,6,19		17				
S53		Open	25,7,26,6, 40,18,19		34				16,17
S54		40	25,7,26,6, 18,19		16				16,17

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* TABLE XVIII. Propagation delay time (t_{PLH}) test - Continued.

Test condition	Parameter	Connect				Initial limit 25°±5°C		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
S55	Propagation delay time (t_{PLH}), V_{DD} = 5 V, C_L = 50 pF	18	25,7,26,6, 40,19	39	37	48	480	ns	16,17
S56	↓	18,40	25,7,26,6,19	39	15	↓	↓	↓	16,17
S57	↓	Open	25,7,26,6, 40,18,39	19	35	↓	↓	↓	15,17
S58	↓	40	25,7,26,6, 18,39	↓	38	↓	↓	↓	15,17
S59	↓	Open	25,7,26,6, 40,18,39	↓	20	↓	↓	↓	15,16
S60	↓	40	25,7,26,6, 18,39	↓	36	↓	↓	↓	15,16
S61	Propagation delay time (t_{PLH}), V_{DD} = 5 V, C_L = 150 pF	Open	7,26,6,40, 18,39,19	25	23	68	680	↓	15,17
S62	↓	Open	7,26,6,40, 18,39,19	↓	27	↓	↓	↓	15,16
S63	↓	7	26,6,40,18, 39,19	↓	4	↓	↓	↓	15,17
S64	↓	7	26,6,40,18, 39,19	↓	8	↓	↓	↓	15,16
S65	↓	26	7,6,40,18, 39,19	↓	24	↓	↓	↓	15,17
S66	↓	26	7,6,40,18, 39,19	↓	2	↓	↓	↓	15,16
S67	↓	,26	6,40,18,39, 19	↓	21	↓	↓	↓	15,17

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* TABLE XVIII. Propagation delay time (t_{PLH}) test - Continued.

Test condition	Parameter	Connect				Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
S68	Propagation delay time (t_{PLH}), V_{DD} = 5 V, C_L = 150 pF	7,26	6,40,18,39,19	25	3	68	680	ns	15,16
S69	↓	6	7,26,40,18,39,19	25	5	↓	↓	↓	15,17
S70		6	7,26,40,18,39,19	25	22				15,16
S71		Open	25,26,6,40,18,39,19	7	23				15,17
S72		25	26,6,40,18,39,19	↓	27				15,17
S73		Open	25,26,6,40,18,39,19		4				15,16
S74		25	26,6,40,18,39,19		8				15,16
S75		26	25,6,40,18,39,19		24				15,17
S76		25,26	6,40,18,39,19		2				15,17
S77		26	25,6,40,18,39,19		21				15,16
S78		25,26	6,40,18,39,19	↓	3				15,16
S79		Open	25,7,6,40,18,39,19	26	23				15,17
S80	↓	25	7,6,40,18,39,19	26	27	↓	↓	↓	15,17

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* TABLE XVIII. Propagation delay time (t_{PLH}) test - Continued.

Test condition	Parameter	Connect				Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
S81	Propagation delay time (t_{PLH}), V_{DD} = 5 V, C_L = 150 pF	7	25,6,40,18, 39,19	26	4	68	680	ns	15,17
S82		7,25	6,40,18,39, 19		8				15,17
S83		Open	25,7,6,40, 18,39,19		24				15,16
S84		25	7,6,40,18, 39,19		2				
S85		7	25,6,40,18, 39,19		21				
S86		7,25	6,40,18,39, 19		3				
S87		Open	25,7,26,40, 18,39,19	6	23				15,17
S88		25	7,26,40,18, 39,19		27				15,17
S89		Open	25,7,26,40, 18,39,19		5				15,16
S90		25	7,26,40,18, 39,19		22				15,16
S91		Open	25,7,26,6, 18,39,19	40	35				15,17
S92		Open	25,7,26,6, 18,39,19		38				15,16
S93		18	25,7,26,6, 39,19		14				15,17

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* TABLE XVIII. Propagation delay time (t_{PLH}) test - Continued.

Test condition	Parameter	Connect				Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
S94	Propagation delay time (t_{PLH}), V_{DD} = 5 V, C_L = 150 pF	18	25,7,26,6, 39,19	40	17	68	680	ns	15,16
S95	↓	39	25,7,26,6, 18,19	↓	34	↓	↓	↓	15,17
S96		39	25,7,26,6, 18,19		16				15,16
S97		18,39	25,7,26,6,19		37				15,17
S98		18,39	25,7,26,6,19		15				15,16
S99		19	25,7,26,6, 18,39		20				15,17
S100		19	25,7,26,6, 18,39		36				15,16
S101		Open	25,7,26,6, 40,39,19	18	35				15,17
S102		40	25,7,26,6, 39,19		38				15,17
S103		Open	25,7,26,6, 40,39,19		14				15,16
S104		40	25,7,26,6, 39,19		17				15,16
S105		39	25,7,26,6, 40,19		34				15,17
S106		39,40	25,7,26,6,19		16				15,17
S107		39	25,7,26,6, 40,19		37				15,16

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* TABLE XVIII. Propagation delay time (t_{PLH}) test - Continued.

Test condition	Parameter	Connect				Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
S108	Propagation delay time (t_{PLH}), V_{DD} = 5 V, C_L = 150 pF	39,40	25,7,26,6,19	18	15	68	680	ns	15,16
S109		Open	25,7,26,6, 40,18,19	39	35				15,17
S110		40	25,7,26,6, 18,19		38				
S111		18	25,7,26,6, 40,19		14				
S112		18,40	25,7,26,6,19		17				
S113		Open	25,7,26,6, 40,18,19		34				15,16
S114		40	25,7,26,6, 18,19		16				
S115		18	25,7,26,6, 40,19		37				
S116		18,40	25,7,26,6,19		15				
S117		Open	25,7,26,6, 40,18,39	19	35				15,17
S118		40	25,7,26,6, 18,39		38				15,17
S119		Open	25,7,26,6, 40,18,39		20				15,16
S120		40	25,7,26,6, 18,39		36				15,16

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* TABLE XIX. Propagation delay time (t_{PHL}) test.

Test condition	Parameter	Connect				Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
T1	Propagation delay time (t_{PHL}), V_{DD} = 5 V, C_L = 50 pF	Open	7 26,6,40, 18,39,19	25	23	48	480	ns	15,17
T2		Open	7,26,6,40, 18,39,19		27				15,16
T3		7	26,6,40,18, 39,19		4				15,17
T4		7	26,6,40,18, 39,19		8				15,16
T5		26	7,6,40,18, 39,19		24				15,17
T6		26	7,6,40,18, 39,19		2				15,16
T7		7,26	6,40,18,39, 19		21				15,17
T8		7,26	6,40,18,39, 19		3				15,16
T9		6	7,26,40,18, 39,19		5				15,17
T10		6	7,26,40,18, 39,19		22				15,16
T11		Open	25,26,6,40, 18,39,19	7	23				15,17
T12		25	26,6,40,18, 39,19		27				15,17
T13		Open	25,26,6,40, 18,39,19		4				15,16

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* TABLE XIX. Propagation delay time (t_{PHL}) test - Continued.

Test condition	Parameter	Connect				Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
T14	Propagation delay time (t_{PHL}), V_{DD} = 5 V, C_L = 50 pF	25	26,6,40,18, 39,19	7	8	48	480	ns	15,16
T15	↓	26	25,6,40,18, 39,19	↓	24	↓	↓	↓	15,17
T16		25,26	6,40,18,39, 19		2				15,17
T17		26	25,6,40,18, 39,19		21				15,16
T18		25,26	6,40,18,39, 19		3				15,16
T19		Open	25,7,6,40, 18,39,19	26	23				15,17
T20		25	7,6,40,18, 39,19	↓	27				↓
T21		7	25,6,40,18, 39,19		4				
T22		7,25	6,40,18,39, 19		8				
T23		Open	25,7,6,40, 18,39,19		24				15,16
T24		25	7,6,40,18, 39,19		2				↓
T25		7	25,6,40,18, 39,19		21				
T26	↓	7,25	6,40,18,39, 19	↓	3	↓	↓	↓	↓

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* TABLE XIX. Propagation delay time (t_{PHL}) test - Continued.

Test condition	Parameter	Connect				Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
T27	Propagation delay time (t_{PHL}), V_{DD} = 5 V, C_L = 50 pF	Open	25,7,26,40, 18,39,19	6	23	48	480	ns	15,17
T28	↓	25	7,26,40,18, 39,19	↓	27	↓	↓	↓	15,17
T29		Open	25,7,26,40, 18,39,19		5				15,16
T30		25	7,26,40,18, 39,19		22				15,16
T31		Open	25,7,26,6, 18,39,19	40	35				15,17
T32		Open	25,7,26,6, 18,39,19		38				15,16
T33		18	25,7,26,6, 39,19		14				15,17
T34		18	25,7,26,6, 39,19		17				15,16
T35		39	25,7,26,6, 18,19		34				15,17
T36		39	25,7,26,6, 18,19		16				15,16
T37		18,39	25,7,26,6,19		37				15,17
T38		18,39	25,7,26,6,19		15				15,16
T39		19	25,7,26,6, 18,39		20				15,17
T40	↓	19	25,7,26,6, 18,39	↓	36	↓	↓	↓	15,16

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* TABLE XIX. Propagation delay time (t_{PHL}) test - Continued.

Test condition	Parameter	Connect				Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
T41	Propagation delay time (t_{PHL}), V_{DD} = 5 V, C_L = 50 pF	Open	25,7,26,6, 40,39,19	18	35	48	480	ns	15,17
T42	↓	40	25,7,26,6, 39,19	↓	38	↓	↓	↓	15,17
T43		Open	25,7,26,6, 40,39,19		14				15,16
T44		40	25,7,26,6, 39,19		17				15,16
T45		39	25,7,26,6, 40,19		34				15,17
T46		39,40	25,7,26,6,19		16				15,17
T47		39	25,7,26,6, 40,19		37				15,16
T48		39,40	25,7,26,6,19		15				15,16
T49		Open	25,7,26,6, 40,18,19	39	35				15,17
T50		40	25,7,26,6, 18,19	↓	38				↓
T51		18	25,7,26,6, 40,19		14				
T52		18,40	25,7,26,6,19						
T53		Open	25,7,26,6, 40,18,19		34				15,16
T54	↓	40	25,7,26,6, 18,19	↓	16	↓	↓	↓	15,16

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* TABLE XIX. Propagation delay time (t_{PHL}) test - Continued.

Test condition	Parameter	Connect				Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
T55	Propagation delay time (t_{PHL}), V_{DD} = 5 V, C_L = 50 pF	18	25,7,26,6, 40,19	39	37	48	480	ns	15,16
T56	↓	18,40	25,7,26,6,19	39	15	↓	↓	↓	15,16
T57	↓	Open	25,7,26,6, 40,18,39	19	35	↓	↓	↓	15,17
T58	↓	40	25,7,26,6, 18,39	↓	38	↓	↓	↓	15,17
T59	↓	Open	25,7,26,6, 40,18,39	↓	20	↓	↓	↓	15,16
T60	↓	40	25,7,26,6, 18,39	↓	36	↓	↓	↓	15,16
T61	Propagation delay time (t_{PHL}), V_{DD} = 5 V, C_L = 150 pF	Open	7,26,6,40, 18,39,19	25	23	68	680	↓	15,17
T62	↓	Open	7,26,6,40, 18,39,19	↓	27	↓	↓	↓	15,16
T63	↓	7	26,6,40,18, 39,19	↓	4	↓	↓	↓	15,17
T64	↓	7	26,6,40,18, 39,19	↓	8	↓	↓	↓	15,16
T65	↓	26	7,6,40,18, 39,19	↓	24	↓	↓	↓	15,17
T66	↓	26	7,6,40,18, 39,19	↓	2	↓	↓	↓	15,16
T67	↓	7,26	6,40,18,39, 19	↓	21	↓	↓	↓	15,17

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* TABLE XIX. Propagation delay time (t_{PHL}) test - Continued.

Test condition	Parameter	Connect				Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
T68	Propagation delay time (t_{PHL}), V_{DD} = 5 V, C_L = 150 pF	7,26	6,40,18,39,19	25	3	68	680	ns	15,16
T69	↓	6	7,26,40,18,39,19	25	5	↓	↓	↓	15,17
T70		6	7,26,40,18,39,19	25	22				15,16
T71		Open	25,26,6,40,18,39,19	7	23				15,17
T72		25	26,6,40,18,39,19	↓	27				15,17
T73		Open	25,26,6,40,18,39,19		4				15,16
T74		25	26,6,40,18,39,19		8				15,16
T75		26	25,6,40,18,39,19		24				15,17
T76		25,26	6,40,18,39,19		2				15,17
T77		26	25,6,40,18,39,19		21				15,16
T78		25,26	6,40,18,39,19	↓	3				15,16
T79		Open	25,7,6,40,18,39,19	26	23				15,17
T80	↓	25	7,6,40,18,39,19	26	27	↓	↓	↓	15,17

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* TABLE XIX. Propagation delay time (t_{PHL}) test - Continued.

Test condition	Parameter	Connect				Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
T81	Propagation delay time (t_{PHL}), V_{DD} = 5 V, C_L = 150 pF	7	25,6,40,18, 39,19	26	4	68	680	ns	15,17
T82	↓	7,25	6,40,18,39, 19	↓	8	↓	↓	↓	15,17
T83		Open	25,7,6,40, 18,39,19		24				15,16
T84		25	7,6,40,18, 39,19		2				↓
T85		7	25,6,40,18, 39,19		21				
T86		7,25	6,40,18,39, 19		3				
T87		Open	25,7,26,40, 18,39,19	6	23				15,17
T88		25	7,26,40,18, 39,19	↓	27				15,17
T89		Open	25,7,26,40, 18,39,19		5				15,16
T90		25	7,26,40,18, 39,19		22				15,16
T91		Open	25,7,26,6, 18,39,19	40	35				15,17
T92		Open	25,7,26,6, 18,39,19	40	38				15,16
T93	↓	18	25,7,26,6, 39,19	40	14	↓	↓	↓	15,17

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* TABLE XIX. Propagation delay time (t_{PHL}) test - Continued.

Test condition	Parameter	Connect				Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
T94	Propagation delay time (t_{PHL}), V_{DD} = 5 V, C_L = 150 pF	18	25,7,26,6, 39,19	40	17	68	680	ns	15,16
T95		39	25,7,26,6, 18,19		34				15,17
T96		39	25,7,26,6, 18,19		16				15,16
T97		18,39	25,7,26,6,19		37				15,17
T98		18,39	25,7,26,6,19		15				15,16
T99		19	25,7,26,6, 18,39		20				15,17
T100		19	25,7,26,6, 18,39		36				15,16
T101		Open	25,7,26,6, 40,39,19	18	35				15,17
T102		40	25,7,26,6, 39,19		38				15,17
T103		Open	25,7,26,6, 40,39,19		14				15,16
T104		40	25,7,26,6, 39,19		17				15,16
T105		39	25,7,26,6, 40,19		34				15,17
T106		39,40	25,7,26,6,19		16				15,17
T107		39	25,7,26,6, 40,19		37				15,16

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* TABLE XIX. Propagation delay time (t_{PHL}) test - Continued.

Test condition	Parameter	Connect				Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
T108	Propagation delay time (t_{PHL}), V_{DD} = 5 V, C_L = 150 pF	39,40	25,7,26,6,19	18	15	68	680	ns	15,16
T109	↓	Open	25,7,26,6, 40,18,19	39	35	↓	↓	↓	15,17
T110	↓	40	25,7,26,6, 16,19	↓	38	↓	↓	↓	↓
T111	↓	18	25,7,26,6, 40,19	↓	14	↓	↓	↓	↓
T112	↓	18,40	25,7,26,6,19	↓	17	↓	↓	↓	↓
T113	↓	Open	25,7,26,6, 40,18,19	↓	34	↓	↓	↓	15,16
T114	↓	40	25,7,26,6, 18,19	↓	16	↓	↓	↓	↓
T115	↓	18	25,7,26,6, 40,19	↓	37	↓	↓	↓	↓
T116	↓	18,40	25,7,26,6,19	↓	15	↓	↓	↓	↓
T117	↓	Open	25,7,26,6, 40,18,39	19	35	↓	↓	↓	15,17
T118	↓	40	25,7,26,6, 18,39	↓	38	↓	↓	↓	15,17
T119	↓	Open	25,7,26,6, 40,18,39	↓	20	↓	↓	↓	15,16
T120	↓	40	25,7,26,6, 18,39	↓	36	↓	↓	↓	15,16

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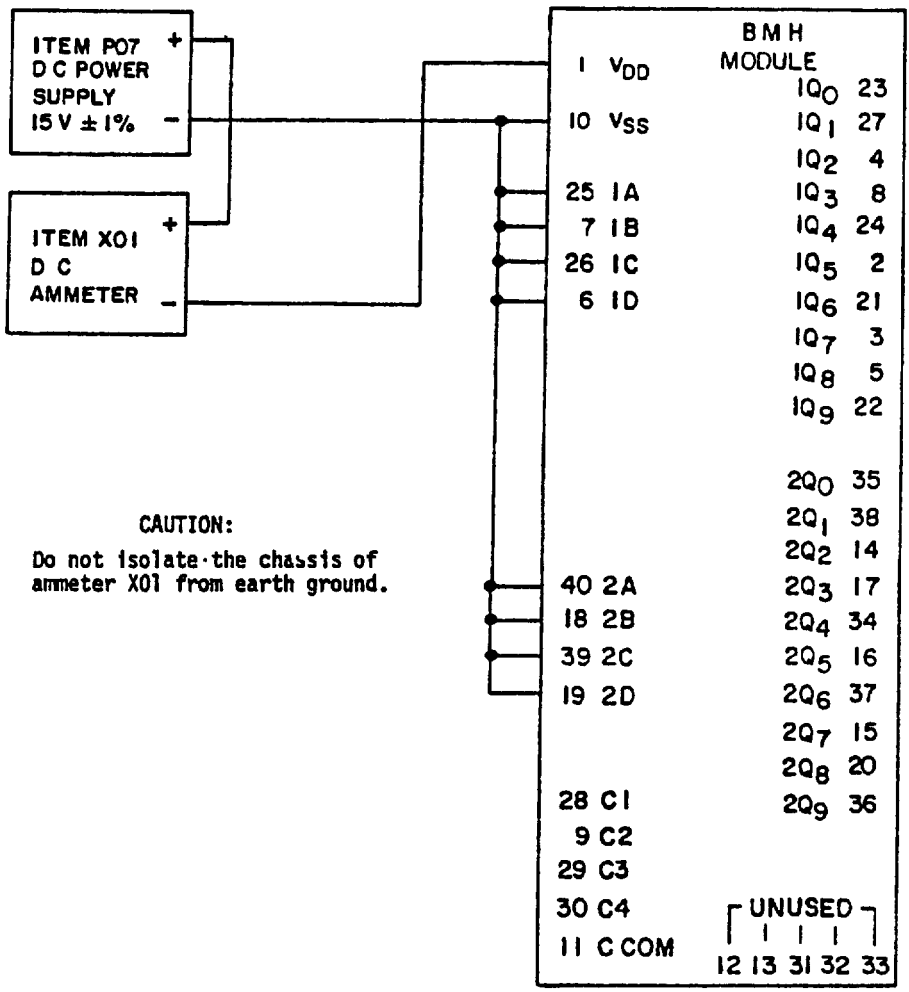
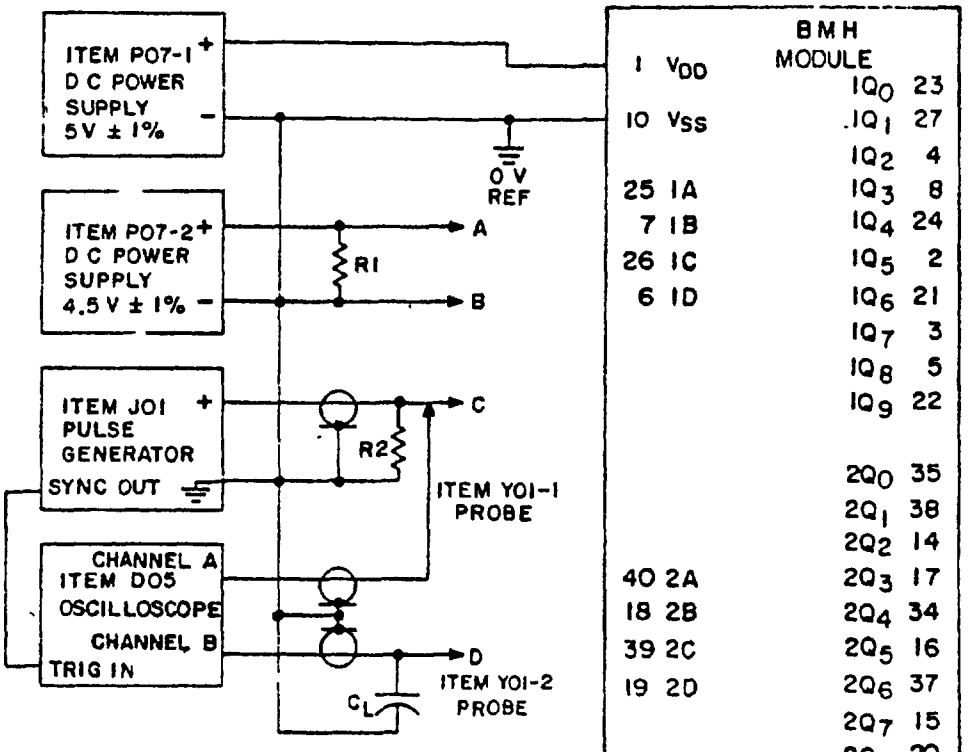


FIGURE 14. Static power supply current (low-level and high-level) test circuit for table XV.

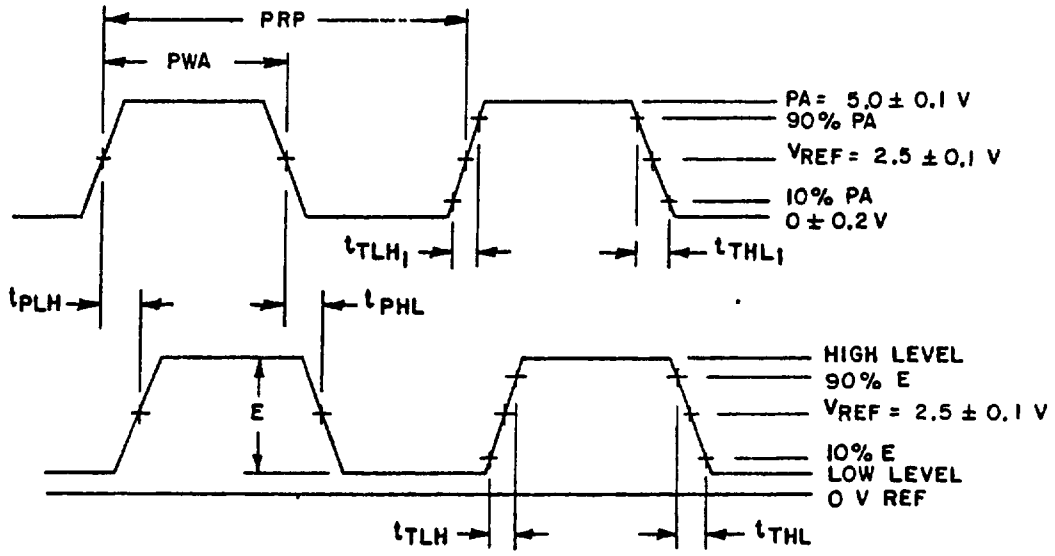
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NOTES:
1. R1 = R2 = 51.1Ω ±5%, 1/2 W.
2. C_L = 50 ±5 pF including probe and parasitic capacitance for test conditions Q1-Q20, R1-R20, S1-S60 and T1-T60.
3. C_L = 150 ±15 pF including probe and parasitic capacitance for test conditions Q21-Q40, R21-R40, S61-S120 and T61-T120.

*FIGURE 15. Transition time and propagation delay time test circuit for tables XVI, XVII, XVIII and XIX.

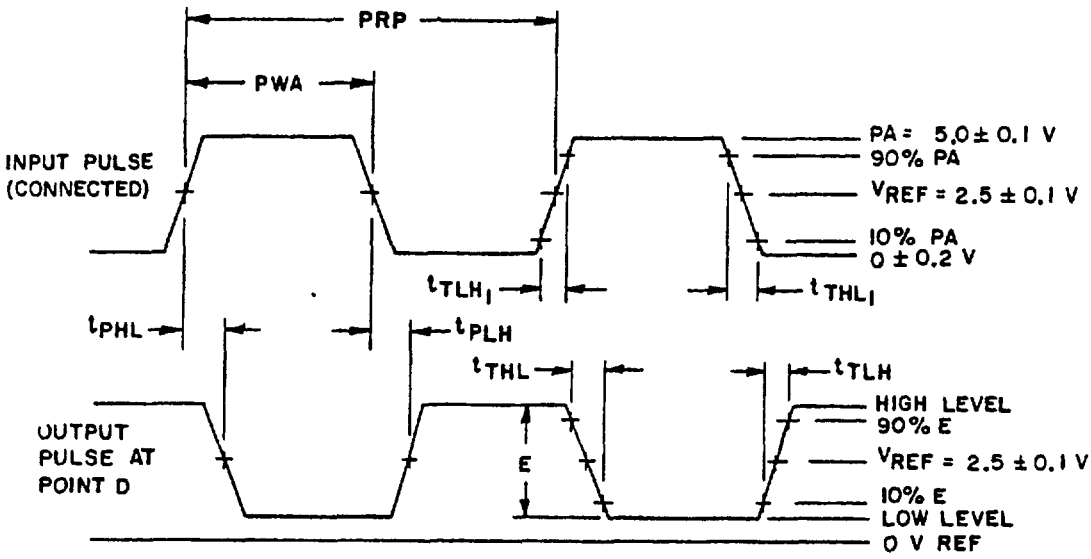
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NOTES:
1. Input pulse parameters:
   $t_{TLH_1} = t_{THL_1} = 10 \pm 2$ ns.
   PWA = 2.5 ±0.2 µs.
   PRP = 5.0 ±0.5 µs.
2. Before making the measurements specified in tables XVI, XVII, XVIII and XIX verify that the output pulse high level is $\geq$ 4.6 Vdc and the low level is $\leq$ 0.4 Vdc.
3. The lead length of electrical conductors between terminator resistor R1, R2 and input pins, probes Y01-1 and Y01-2 to input and output pins and between load circuit and output pins shall be less than 1.00 inch (25.4 mm).
4. All voltages are referenced to V_{SS}.

* FIGURE 16. Transition time and propagation delay time test circuit waveforms for tables XVI, XVII, XVIII and XIX.

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NOTES:
1. Input pulse parameters:
   $t_{TLH_1} = t_{THL_1} = 10 \pm 2$ ns.
   PWA = 2.5 ±0.2 µs.
   PRP = 5.0 ±0.5 µs.
2. Before making the measurements specified in tables XVI, XVII, XVIII and XIX verify that the output pulse high level is $\geq$ 4.6 V and the low level is $\leq$ 0.4 V.
3. The lead length of electrical conductors between terminator resistor R1, R2 and input pins, probes Y01-1 and Y01-2 to input and output pins and between load circuit and output pins shall be less than 1.00 inch (25.4 mm).
4. All voltages are referenced to V_{SS}.

*FIGURE 17. Transition time and propagation delay time test circuit waveforms for tables XVI, XVII, XVIII and XIX.

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TABLE XX. Transition time (t_{TLH}) test.

Test condition	Parameter	Connect				Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
U1	Transition time (t_{TLH}), V_{DD} = 10 V, C_L = 50 pF	Open	7,26,6,40, 18,39,19	25	23	15	150	ns	18,20
U2		Open	7,26,6,40, 18,39,19	25	27				18,19
U3		Open	25,26,6,40, 18,39,19	7	4				
U4		25	26,6,40,18, 39,19	7	8				
U5		Open	25,7,6,40, 18,39,19	26	24				
U6		25	7,6,40,18, 39,19		2				
U7		7	25,6,40,18, 39,19		21				
U8		25,7	6,40,18,39, 19		3				
U9		Open	25,7,26,40, 18,39,19	6	5				
U10		25	7,26,40,18, 39,19	6	22				
U11		Open	25,7,26,6, 18,39,19	40	35				18,20
U12		Open	25,7,26,6, 18,39,19	40	38				18,19
U13		Open	25,7,26,6, 40,39,19	18	14				18,19

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TABLE XX. Transition time (t_{TLH}) test - Continued.

Test condition	Parameter	Connect				Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
U14	Transition time (t_{TLH}), V_{DD} = 10 V, C_L = 50 pF	40	25,7,26,6, 39,19	18	17	15	150	ns	18,19
U15	↓	Open	25,7,26,6, 40,18,19	39	34	↓	↓	↓	↓
U16		40	25,7,26,6, 18,19	↓	16				
U17		18	25,7,26,6, 40,19		37				
U18		40,18	25,7,26,6,19		15				
U19		Open	25,7,26,6, 40,18,39	19	20				
U20		40	25,7,26,6, 18,39	19	36				

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TABLE XXI. Transition time (t_{THL}) test.

Test condition	Parameter	Connect				Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
V1	Transition time (t_{THL}), V_{DD} = 10 V, C_L = 50 pF	Open	7,26,6,40, 18,39,19	25	23	7	75	ns	18,20
V2	↓	Open	7,26,6,40, 18,39,19	25	27	↓	↓	↓	18,19
V3	↓	Open	25,26,6,40, 18,39,19	7	4	↓	↓	↓	↓
V4	↓	25	26,6,40,18, 39,19	7	8	↓	↓	↓	↓
V5	↓	Open	25,7,6,40, 18,39,19	26	24	↓	↓	↓	↓
V6	↓	25	7,6,40,18, 39,19	↓	2	↓	↓	↓	↓
V7	↓	7	25,6,40,18, 39,19	↓	21	↓	↓	↓	↓
V8	↓	25,7	6,40,18,39, 19	↓	3	↓	↓	↓	↓
V9	↓	Open	25,7,26,40, 18,39,19	6	5	↓	↓	↓	↓
V10	↓	25	7,26,40,18, 39,19	6	22	↓	↓	↓	↓
V11	↓	Open	25,7,26,6, 18,39,19	40	35	↓	↓	↓	18,20
V12	↓	Open	25,7,26,6, 18,39,19	40	38	↓	↓	↓	18,19
V13	↓	Open	25,7,26,6, 40,39,19	18	14	↓	↓	↓	18,19

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TABLE XXI. Transition time (t_{THL}) test - Continued.

Test condition	Parameter	Connect				Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
V14	Transition time (t_{THL}), V_{DD} = 10 V, C_L = 50 pF	40	25,7,26,6, 39,19	16	17	7	75	ns	18,19
V15	↓	Open	25,7,26,6, 40,18,19	39	34	↓	↓	↓	↓
V16		40	25,7,26,6, 18,19	↓	16				
V17		18	25,7,26,6, 40,19		37				
V18		40,18	25,7,26,6,19		15				
V19		Open	25,7,26,6, 40,18,39	19	20				
V20		40	25,7,26,6, 18,39	19	36				

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* TABLE XXII. Propagation delay time (t_{PLH}) test.

Test condition	Parameter	Connect				Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
W1	Propagation delay time (t_{PLH}), V_{DD} = 10 V, C_L = 50 pF	Open	7,26,6,40, 18,39,19	25	23	20	200	ns	18,20
W2	↓	Open	7,26,6,40, 18,39,19	↓	27	↓	↓	↓	18,19
W3	↓	7	26,6,40,18, 39,19	↓	4	↓	↓	↓	18,20
W4	↓	7	26,6,40,18, 39,19	↓	8	↓	↓	↓	18,19
W5	↓	26	7,6,40,18, 39,19	↓	24	↓	↓	↓	18,20
W6	↓	26	7,6,40,18, 39,19	↓	2	↓	↓	↓	18,19
W7	↓	7,26	6,40,18,39, 19	↓	21	↓	↓	↓	18,20
W8	↓	7,26	6,40,18,39, 19	↓	3	↓	↓	↓	18,19
W9	↓	6	7,26,40,18, 39,19	↓	5	↓	↓	↓	18,20
W10	↓	6	7,26,40,18, 39,19	↓	20	↓	↓	↓	18,19
W11	↓	Open	25,26,6,40, 18,39,19	7	23	↓	↓	↓	18,20
W12	↓	25	26,6,40,18, 39,19	↓	27	↓	↓	↓	18,20
W13	↓	Open	25,26,6,40, 18,39,19	↓	4	↓	↓	↓	18,19

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* TABLE XXII. Propagation delay time (t_{PLH}) test - Continued.

Test condition	Parameter	Connect				Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
W14	Propagation delay time (t_{PLH}), V_{DD} = 10 V, C_L = 50 pF	25	26,6,40,18, 39,19	7	8	20	200	ns	18,19
W15		26	25,6,40,18, 39,19		24				18,20
W16		25,26	6,40,18,39, 19		2				18,20
W17		26	25,6,40,18, 39,19		21				18,19
W18		25,26	6,40,18,39, 19		3				18,19
W19		Open	25,7,6,40, 18,39,19	26	23				18,20
W20		25	7,6,40,18, 39,19		27				
W21		7	25,6,40,18, 39,19		4				
W22		7,25	6,40,18,39, 19		8				
W23		Open	25,7,6,40, 18,39,19		24				18,19
W24		25	7,6,40,18, 39,19		2				
W25		7	25,6,40,18, 39,19		21				
W26		7,25	6,40,18,39, 19		3				

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* TABLE XXII. Propagation delay time (t_{PLH}) test - Continued.

Test condition	Parameter	Connect				Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
W27	Propagation delay time (t_{PLH}), V_{DD} = 10 V, C_L = 50 pF	Open	25,7,26,40, 18,39,19	6	23	20	200	ns	18,20
W28		25	7,26,40,18, 39,19		27				18,20
W29		Open	25,7,26,40, 18,39,19		5				18,19
W30		25	7,26,40,18, 39,19		22				18,19
W31		Open	25,7,26,6, 18,39,19	40	35				18,20
W32		Open	25,7,26,6, 18,39,19		38				18,19
W33		18	25,7,26,6, 39,19		14				18,20
W34		18	25,7,26,6, 39,19		17				18,19
W35		39	25,7,26,6, 18,19		34				18,20
W36		39	25,7,26,6, 18,19		16				18,19
W37		18,39	25,7,26,6,19		37				18,20
W38		18,39	25,7,26,6,19		15				18,19
W39		19	25,7,26,6, 18,39		20				18,20
W40		19	25,7,26,6, 18,39		36				18,19

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* TABLE XXII. Propagation delay time (t_{PLH}) test - Continued.

Test condition	Parameter	Connect A to pin	Connect B to pin	Connect C to pin	Connect D to pin	Initial limits 25°±5°C ambient Min	Initial limits 25°±5°C ambient Max	Unit	Test figure
W41	Propagation delay time (t_{PLH}), V_{DD} = 10 V, C_L = 50 pF	Open	25,7,26,6, 40,39,19	18	35	20	200	ns	18,20
W42		40	25,7,26,6, 39,19		38				18,20
W43		Open	25,7,26,6, 40,39,19		14				18,19
W44		40	25,7,26,6, 39,19		17				18,19
W45		39	25,7,26,6, 40,19		34				18,20
W46		39,40	25,7,26,6,19		16				18,20
W47		39	25,7,26,6, 40,19		37				18,19
W48		39,40	25,7,26,6,19		15				18,19
W49		Open	25,7,26,6, 40,18,19	39	35				18,20
W50		40	25,7,26,6, 18,19		38				
W51		18	25,7,26,6, 40,19		14				
W52		18,40	25,7,26,6,19		17				
W53		Open	25,7,26,6, 40,18,19		34				18,19
W54		40	25,7,26,6, 18,19		16				18,19

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* TABLE XXII. Propagation delay time (t_{PLH}) test - Continued.

Test condition	Parameter	Connect				Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
W55	Propagation delay time (t_{PLH}), V_{DD} = 10 V, C_L = 50 pF	18	25,7,26,6, 40,19	39	37	20	200	ns	18,19
W56	↓	18,40	25,7,26,6,19	39	15	↓	↓	↓	18,19
W57	↓	Open	25,7,26,6, 40,18,39	19	35	↓	↓	↓	18,20
W58	↓	40	25,7,26,6, 18,39	↓	38	↓	↓	↓	18,20
W59	↓	Open	25,7,26,6, 40,18,39	↓	20	↓	↓	↓	18,19
W60	↓	40	25,7,26,6, 18,39	↓	36	↓	↓	↓	18,19

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* TABLE XXIII. Propagation delay time (t_{PHL}) test.

Test condition	Parameter	Connect				Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
X1	Propagation delay time (t_{PHL}), V_{DD} = 10 V, C_L = 50 pF	Open	7,26,6,40, 18,39,19	25	23	20	200	ns	18,20
X2	↓	Open	7,26,6,40, 18,39,19	↓	27	↓	↓	↓	18,19
X3	↓	7	26,6,40,18, 39,19	↓	4	↓	↓	↓	18,20
X4	↓	7	26,6,40,18, 39,19	↓	8	↓	↓	↓	18,19
X5	↓	26	7,6,40,18, 39,19	↓	24	↓	↓	↓	18,20
X6	↓	26	7,6,40,18, 39,19	↓	2	↓	↓	↓	18,19
X7	↓	7,26	6,40,18,39, 19	↓	21	↓	↓	↓	18,20
X8	↓	7,26	6,40,18,39, 19	↓	3	↓	↓	↓	18,19
X9	↓	6	7,26,40,18, 39,19	↓	5	↓	↓	↓	18,20
X10	↓	6	7,26,40,18, 39,19	↓	22	↓	↓	↓	18,19
X11	↓	Open	25,26,6,40, 18,39,19	7	23	↓	↓	↓	18,20
X12	↓	25	26,6,40,18, 39,19	↓	27	↓	↓	↓	18,19
X13	↓	Open	25,26,6,40, 18,39,19	↓	4	↓	↓	↓	18,20

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* TABLE XXIII. Propagation delay time (t_{PHL}) test - Continued.

Test condition	Parameter	Connect				Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
X14	Propagation delay time (t_{PHL}), V_{DD} = 10 V, C_L = 50 pF	25	26,6,40,18, 39,19	7	8	20	200	ns	18,19
X15		26	25,6,40,18, 39,19		24				18,20
X16		25,26	6,40,18,39, 19		2				18,20
X17		26	25,6,40,18, 39,19		21				18,19
X18		25,26	6,40,18,39, 19		3				18,19
X19		Open	25,7,6,40, 18,39,19	26	23				18,20
X20		25	7,6,40,18, 39,19		27				
X21		7	25,6,40,18, 39,19		4				
X22		7,25	6,40,18,39, 19		8				
X23		Open	25,7,6,40, 18,39,19		24				18,19
X24		25	7,6,40,18, 39,19		2				
X25		7	25,6,40,18, 39,19		21				
X26		7,25	6,40,18,39, 19		3				

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* TABLE XXIII. Propagation delay time (t_{PHL}) test - Continued.

Test condition	Parameter	Connect				Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
X27	Propagation delay time (t_{PHL}), V_{DD} = 10 V, C_L = 50 pF	Open	25,7,26,40, 18,39,19	6	23	20	200	ns	18,20
X28		25	7,26,40,18, 39,19		27				18,20
X29		Open	25,7,26,40, 18,39,19		5				18,19
X30		25	7,26,40,18, 39,19		22				18,19
X31		Open	25,7,26,6, 18,39,19	40	35				18,20
X32		Open	25,7,26,6, 18,39,19		38				18,19
X33		18	25,7,26,6, 39,19		14				18,20
X34		18	25,7,26,6, 39,19		17				18,19
X35		39	25,7,26,6, 18,19		34				18,20
X36		39	25,7,26,6, 18,19		16				18,19
X37		18,39	25,7,26,6,19		37				18,20
X38		18,39	25,7,26,6,19		15				18,19
X39		19	25,7,26,6, 18,39		20				18,20
X40		19	25,7,26,6, 18,39		36				18,19

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* TABLE XXIII. Propagation delay time (t_{PHL}) test - Continued.

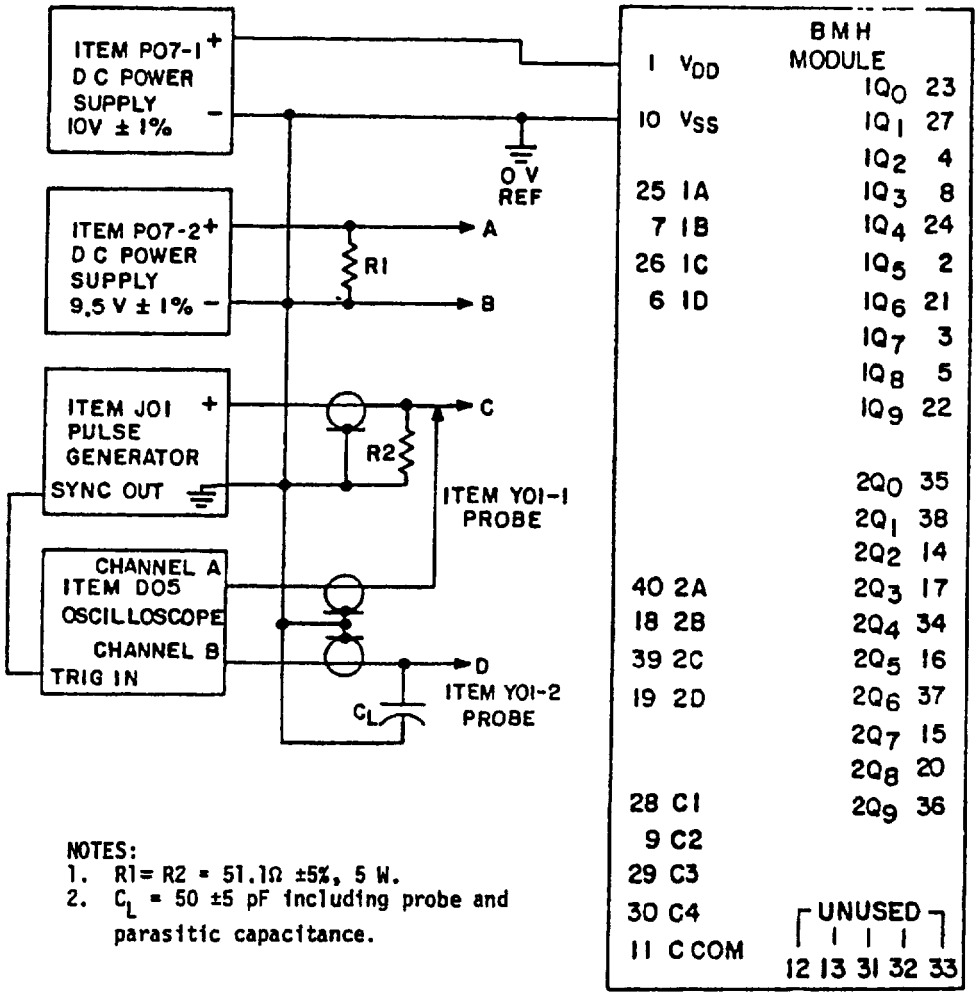
Test condition	Parameter	Connect				Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
X41	Propagation delay time (t_{PHL}), V_{DD} = 10 V, C_L = 50 pF	Open	25,7,26,6, 40,39,19	18	35	20	200	ns	18,20
X42		40	25,7,26,6, 39,19		38				18,19
X43		Open	25,7,26,6, 40,39,19		14				18,20
X44		40	25,7,26,6, 39,19		17				18,19
X45		39	25,7,26,6, 40,19		34				18,20
X46		39,40	25,7,26,6,19		16				18,20
X47		39	25,7,26,6, 40,19		37				18,19
X48		39,40	25,7,26,6,19		15				18,19
X49		Open	25,7,26,6, 40,18,19	39	35				18,20
X50		40	25,7,26,6, 18,19		38				
X51		18	25,7,26,6, 40,19		14				
X52		18,40	25,7,26,6,19		17				
X53		Open	25,7,26,6, 40,18,19		34				18,19
X54		40	25,7,26,6, 18,19		16				18.19

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* TABLE XXIII. Propagation delay time (t_{PHL}) test - Continued.

Test condition	Parameter	Connect				Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
X55	Propagation delay time (t_{PHL}), V_{DD} = 10 V, C_L = 50 pF	18	25,7,26,6, 40,19	39	37	20	200	ns	18,19
X56	↓	18,40	25,7,26,6,19	39	15	↓	↓	↓	18,19
X57	↓	Open	25,7,26,6, 40,18,39	19	35	↓	↓	↓	18,20
X58	↓	40	25,7,26,6, 18,39	↓	38	↓	↓	↓	18,20
X59	↓	Open	25,7,26,6, 40,18,39	↓	20	↓	↓	↓	18,19
X60	↓	40	25,7,26,6, 18,39	↓	36	↓	↓	↓	18,19

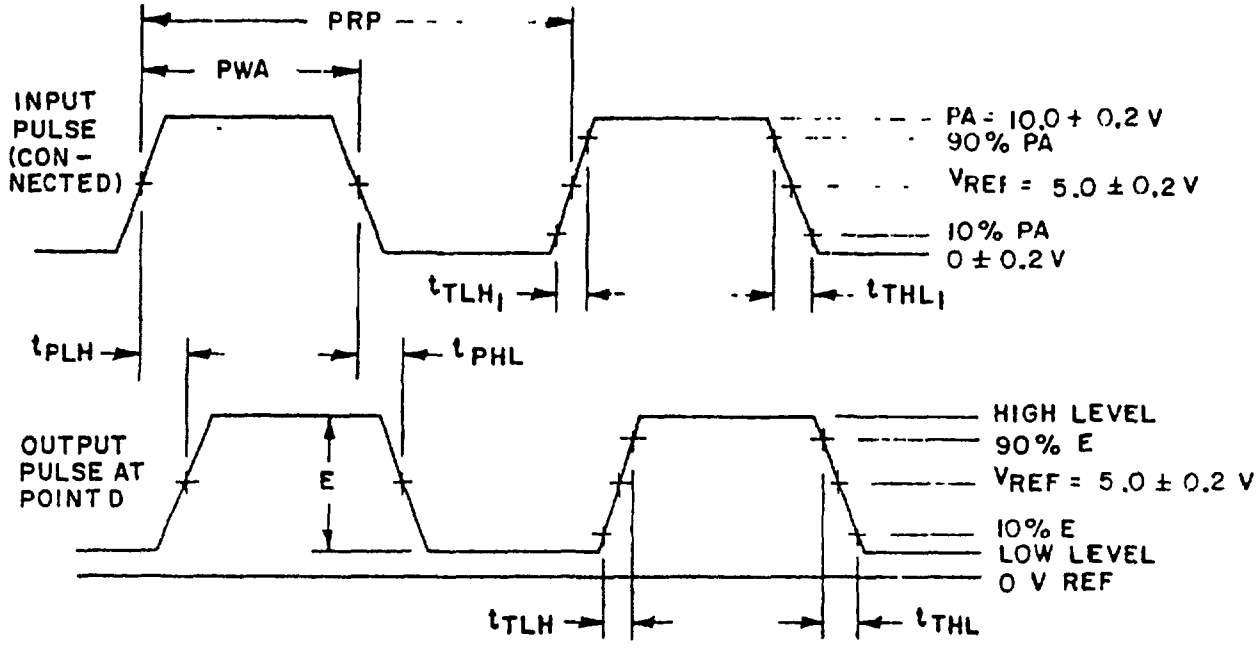
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NOTES:
1. R1 = R2 = 51.1Ω ±5%, 5 W.
2. C_L = 50 ±5 pF including probe and parasitic capacitance.

FIGURE 18. Transition time and propagation delay time test circuit for tables XX, XXI, XXII and XXIII.

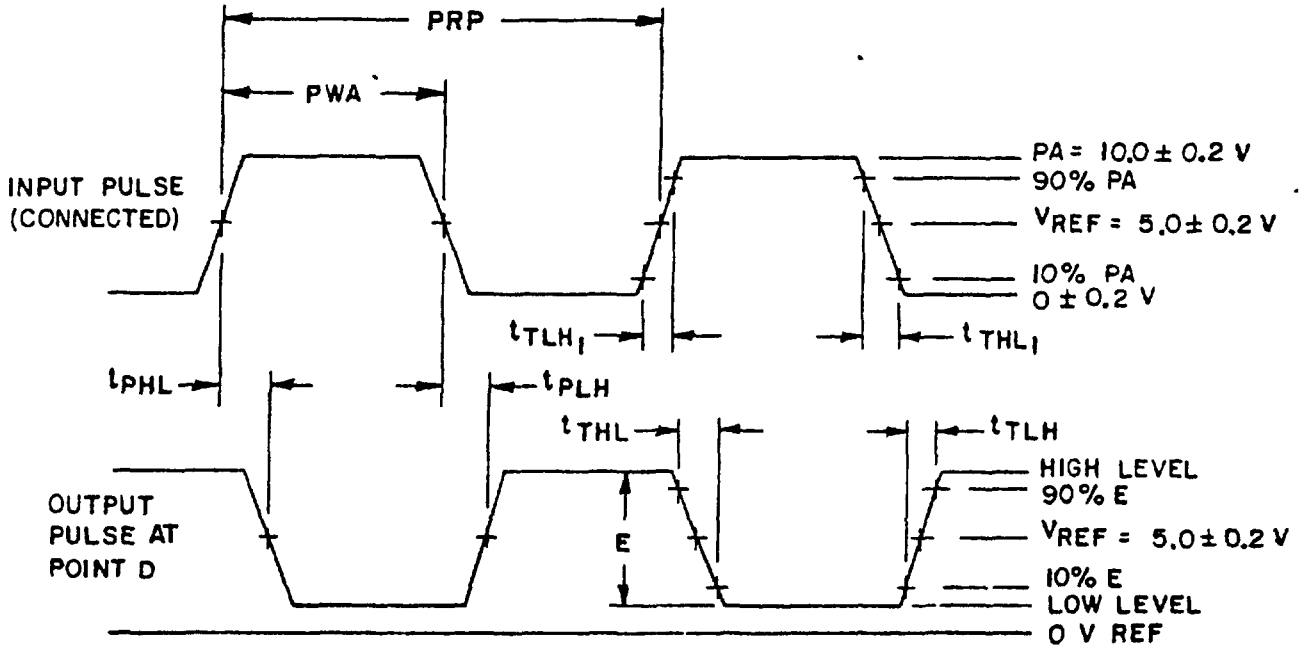
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NOTES:
1. Input pulse parameters:
   $t_{TLH_1} = t_{THL_1} = 10 \pm 2$ ns.
   PWA = 2.5 ±0.2 µs.
   PRP = 5.0 ±0.5 µs.
2. Before making the measurements specified in tables XX, XXI, XXII and XXIII verify that the output pulse high level is $\geq$ 9.5 V and the low level is $\leq$ 0.5 V.
3. The lead length of electrical conductors between terminator resistor R1, R2 and input pins, probes YO1-1 and YO1-2 to input and output pins and between load circuit and output pins shall be less than 1.00 inch (25.4 mm).
4. All voltages are referenced to V_{SS}.

FIGURE 19. Transition time and propagation delay time test circuit waveforms for tables XX, XXI, XXII and XXIII.

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NOTES:
1. Input pulse parameters:
   $t_{TLH_1} = t_{THL_1} = 10 \pm 2$ ns.
   PWA = 2.5 ±0.2 µs.
   PRP = 5.0 ±0.5 µs.
2. Before making the measurements specified in tables XX, XXI, XXII and XXIII verify that the output pulse high level is $\geq$ 9.5 V and the low level is $\leq$ 0.5 V.
3. The lead length of electrical conductors between terminator resistor R1, R2 and input pins, probes YO1-1 and YO1-2 to input and output pins and between load circuit and output pin shall be less than 1.00 inch (25.4 mm).
4. All voltages are referenced to V_{SS}.

FIGURE 20. Transition time and propagation delay time test circuit waveforms for tables XX, XXI, XXII and XXIII.

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TABLE XXIV. Transition time (t_{TLH}) test.

Test condition	Parameter	Connect				Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
Y1	Transition time (t_{TLH}), V_{DD} = 15 V, C_L = 50 pF	Open	7,26,6,40, 18,39,19	25	23	11	110	ns	21,23
Y2		Open	7,26,6,40, 18,39,19	25	27				21,22
Y3		Open	25,26,6,40, 18,39,19	7	4				
Y4		25	26,6,40,18, 39,19	7	8				
Y5		Open	25,7,6,40, 18,39,19	26	24				
Y6		25	7,6,40,18, 39,19		2				
Y7		7	25,6,40,18, 39,19		21				
Y8		25,7	6,40,18,39, 19		3				
Y9		Open	25,7,26,40, 18,39,19	6	5				
Y10		25	7,26,40,18, 39,19	6	22				
Y11		Open	25,7,26,6, 18,39,19	40	35				21,23
Y12		Open	25,7,26,6, 18,39,19	40	38				21,22
Y13		Open	25,7,26,6, 40,39,19	18	14				21,22

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TABLE XXIV. Transition time (t_{TLH}) test - Continued.

Test condition	Parameter	Connect				Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
Y14	Transition time (t_{TLH}), V_{DD} = 15 V, C_L = 50 pF	40	25,7,26,6, 39,19	18	17	11	110	ns	21,22
Y15	↓	Open	25,7,26,6, 40,18,19	39	34	↓	↓	↓	↓
Y16		40	25,7,26,6, 18,19	↓	16				
Y17		18	25,7,26,6, 40,19		37				
Y18		40,18	25,7,26,6,19		15				
Y19		Open	25,7,26,6, 40,18,39	19	20				
Y20	↓	40	25,7,26,6, 18,39	19	36	↓	↓	↓	↓

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TABLE XXV. Transition time (t_{THL}) test.

Test condition	Parameter	Connect				Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
Z1	Transition time (t_{THL}), V_{DD} = 15 V, C_L = 50 pF	Open	7,26,6,40, 18,39,19	25	23	5	55	ns	21,23
Z2		Open	7,26,6,40, 18,39,19	25	27				21,22
Z3		Open	25,26,6,40, 18,39,19	7	4				
Z4		25	26,6,40,18, 39,19	7	8				
Z5		Open	25,7,6,40, 18,39,19	26	24				
Z6		25	7,6,40,18, 39,19		2				
Z7		7	25,6,40,18, 39,19		21				
Z8		25,7	6,40,18,39, 19		3				
Z9		Open	25,7,26,40, 18,39,19	6	5				
Z10		25	7,26,40,18, 39,19	6	22				
Z11		Open	25,7,26,6, 18,39,19	40	35				21,23
Z12		Open	25,7,26,6, 18,39,19	40	38				21,22
Z13		Open	25,7,26,6, 40,39,19	18	14				21,22

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TABLE XXV. Transition time (t_{THL}) test - Continued.

Test condition	Parameter	Connect				Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
Z14	Transition time (t_{THL}), V_{DD} = 15 V, C_L = 50 pF	40	25,7,26,6, 39,19	18	17	5	55	ns	21,22
Z15	↓	Open	25,7,26,6, 40,18,19	39	34	↓	↓	↓	↓
Z16		40	25,7,26,6, 18,19	↓	16				
Z17		18	25,7,26,6, 40,19		37				
Z18		40,18	25,7,26,6,19		15				
Z19		Open	25,7,26,6, 40,18,39	19	20				
Z20		40	25,7,26,6, 18,39	19	36				

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* TABLE XXVI. Propagation delay time (t_{PLH}) test.

Test condition	Parameter	Connect				Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
AA1	Propagation delay time (t_{PLH}), V_{DD} = 15 V, C_L = 50 pF	Open	7,26,6,40, 18,39,19	25	23	14	140	ns	21,23
AA2		Open	7,26,6,40, 18,39,19		27				21,22
AA3		7	26,6,40,18, 39,19		4				21,23
AA4		7	26,6,40,18, 39,19		8				21,22
AA5		26	7,6,40,18, 39,19		24				21,23
AA6		26	7,6,40,18, 39,19		2				21,22
AA7		7,26	6,40,18,39, 19		21				21,23
AA8		7,26	6,40,18,39, 19		3				21,22
AA9		6	7,26,40,18, 39,19		5				21,23
AA10		6	7,26,40,18, 39,19		22				21,22
AA11		Open	25,26,6,40, 18,39,19	7	23				21,23
AA12		25	26,6,40,18, 39,19	7	27				21,23
AA13		Open	25,26,6,40, 18,39,19	7	4				21,22

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* TABLE XXVI. Propagation delay time (t_{PLH}) test - Continued.

Test condition	Parameter	Connect A to pin	Connect B to pin	Connect C to pin	Connect D to pin	Initial limits 25°±5°C ambient Min	Initial limits 25°±5°C ambient Max	Unit	Test figure
AA14	Propagation delay time (t_{PLH}), V_{DD} = 15 V, C_L = 50 pF	25	26,6,40,16, 39,19	7	8	14	140	ns	21,22
AA15	↓	26	25,6,40,18, 39,19	↓	24	↓	↓	↓	21,23
AA16	↓	25,26	6,40,18,39, 19	↓	2	↓	↓	↓	21,23
AA17	↓	26	25,6,40,18, 39,19	↓	21	↓	↓	↓	21,22
AA18	↓	25,26	6,40,18,39, 19	↓	3	↓	↓	↓	21,22
AA19	↓	Open	25,7,6,40, 18,39,19	26	23	↓	↓	↓	21,23
AA20	↓	25	7,6,40,18, 39,19	↓	27	↓	↓	↓	↓
AA21	↓	7	25,6,40,18, 39,19	↓	4	↓	↓	↓	↓
AA22	↓	7,25	6,40,18,39, 19	↓	8	↓	↓	↓	↓
AA23	↓	Open	25,7,6,40, 18,39,19	↓	24	↓	↓	↓	21,22
AA24	↓	25	7,6,40,16, 39,19	↓	2	↓	↓	↓	↓
AA25	↓	7	25,6,40,18, 39,19	↓	21	↓	↓	↓	↓
AA26	↓	7,25	6,40,18,39, 19	↓	3	↓	↓	↓	↓

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* TABLE XXVI. Propagation delay time (t_{PLH}) test - Continued.

Test condition	Parameter	Connect				Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
AA27	Propagation delay time (t_{PLH}), V_{DD} = 15 V, C_L = 50 pF	Open	25,7,26,40, 18,39,19	6	23	14	140	ns	21,23
AA28	↓	25	7,26,40,18, 39,19	↓	27	↓	↓	↓	21,23
AA29	↓	Open	25,7,26,40, 18,39,19	↓	5	↓	↓	↓	21,22
AA30	↓	25	7,26,40,18, 39,19	↓	22	↓	↓	↓	21,22
AA31	↓	Open	25,7,26,6, 18,39,19	40	35	↓	↓	↓	21,23
AA32	↓	Open	25,7,26,6, 18,39,19	↓	38	↓	↓	↓	21,22
AA33	↓	16	25,7,26,6, 39,19	↓	14	↓	↓	↓	21,23
AA34	↓	18	25,7,26,6, 39,19	↓	17	↓	↓	↓	21,22
AA35	↓	39	25,7,26,6, 18,19	↓	34	↓	↓	↓	21,23
AA36	↓	39	25,7,26,6, 18,19	↓	16	↓	↓	↓	21,22
AA37	↓	18,39	25,7,26,6,19	↓	37	↓	↓	↓	21,23
AA38	↓	18,39	25,7,26,6,19	↓	15	↓	↓	↓	21,22
AA39	↓	19	25,7,26,6, 18,39	↓	20	↓	↓	↓	21,23
AA40	↓	19	25,7,26,6, 18,39	↓	36	↓	↓	↓	21,22

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* TABLE XXVI. Propagation delay time (t_{PLH}) test - Continued.

Test condition	Parameter	Connect				Initial limits 25 ±5°C ambient		Unit	Test. figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
AA41	Propagation delay time (t_{PLH}), V_{DD} = 15 V, C_L = 50 pF	Open	25,7,26,6, 40,39,19	18	35	14	140	ns	21,23
AA42		40	25,7,26,6, 39,19		38				21,23
AA43		Open	25,7,26,6, 40,39,19		14				21,22
AA44		40	25,7,26,6, 39,19		17				21,22
AA45		39	25,7,26,6, 40,19		34				21,23
AA46		39,40	25,7,26,6,19		16				21,23
AA47		39	25,7,26,6, 40,19		37				21,22
AA48		39,40	25,7,26,6,19		15				21,22
AA49		Oper	25,7,26,6, 40,18,19	39	35				21,23
AA50		40	25,7,26,6, 18,19		38				
AA51		18	25,7,26,6, 40,19		14				
AA52		18,40	25,7,26,6,19		17				
AA53		Open	25,7,26,6, 40,18,19		34				21,22
AA54		40	25,7,26,6, 18,19		16				21,22

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* TABLE XXVI. Propagation delay time (t_{PLH}) test - Continued.

Test condition	Parameter	Connect				Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
AA55	Propagation delay time (t_{PLH}), V_{DD} = 15 V, C_L = 50 pF	18	25,7,26,6, 40,19	39	37	14	140	ns	21,22
AA56	↓	18,40	25,7,26,6,19	39	15	↓	↓	↓	21,22
AA57	↓	Open	25,7,26,6, 40,18,39	19	35	↓	↓	↓	21,23
AA58	↓	40	25,7,26,6, 18,39	↓	38	↓	↓	↓	21,23
AA59	↓	Open	25,7,26,6, 40,18,39	↓	20	↓	↓	↓	21,22
AA60	↓	40	25,7,26,6, 18,39	↓	36	↓	↓	↓	21,22

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* TABLE XXVII. Propagation delay time (t_{PHL}) test.

Test condition	Parameter	Connect				Initial limits 25°±5°C ambient		Unit	Test-figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
BB1	Propagation delay time (t_{PHL}), V_{DD} = 15 V, C_L = 50 pF	Open	7,26,6,40, 18,39,19	25	23	14	140	ns	21,23
BB2	↓	Open	7,26,6,40, 18,39,19	↓	27	↓	↓	↓	21,22
BB3		7	26,6,40,18, 39,19		4				21,23
BB4		7	26,6,40,18, 39,19		8				21,22
BB5		26	7,6,40,18, 39,19		24				21,23
BB6		26	7,6,40,18, 39,19		2				21,22
BB7		7,26	6,40,18,39, 19		21				21,23
BB8		7,26	6,40,18,39, 19		3				21,22
BB9		6	7,26,40,18, 39,19		5				21,23
BB10		6	7,26,40,18, 39,19		22				21,22
BB11		Open	25,26,6,40, 18,39,19	7	23				21,23
BB12		25	26,6,40,18, 39,19	7	27				21,23
BB13	↓	Open	25,26,6,40, 18,39,19	7	4	↓	↓	↓	21,22

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* TABLE XXVII. Propagation delay time (t_{PHL}) test - Continued.

Test condition	Parameter	Connect				Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
BB14	Propagation delay time (t_{PHL}), V_{DD} = 15 V, C_L = 50 pF	25	26,6,40,18, 39,19	7	8	14	140	ns	21,22
BB15		26	25,6,40,18, 39,19		24				21,23
BB16		25,26	6,40,18,39, 19		2				21,23
BB17		26	25,6,40,18, 39,19		21				21,22
BB18		25,26	6,40,18,39, 19		3				21,22
BB19		Open	25,7,6,40, 18,39,19	26	23				21,23
BB20		25	7,6,40,18, 39,19		27				
BB21		7	25,6,40,18, 39,19		4				
BB22		7,25	6,40,18,39, 19		8				
BB23		Open	25,7,6,40, 18,39,19		24				21,22
BB24		25	7,6,40,18, 39,19		2				
BB25		7	25,6,40,18, 39,19		21				
BB26		7,25	6,40,18,39, 19		3				

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* TABLE XXVII. Propagation delay time (t_{PHL}) test - Continued.

Test condition	Parameter	Connect				Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
BB27	Propagation delay time (t_{PHL}), V_{DD} = 15 V, C_L = 50 pF	Open	25,7,26,40, 18,39,19	6	23	14	140	ns	21,23
BB28		25	7,26,40,18 39,19		27				21,23
BB29		Oper	25,7,26,40, 18,39,19		5				21,22
BB30		25	7,26,40,18, 39,19		22				21,22
BB31		Open	25,7,26,6, 18,39,19	40	35				21,23
BB32		Open	25,7,26,6, 18,39,19		38				21,22
BB33		18	25,7,26,6, 39,19		14				21,23
BB34		18	25,7,26,6, 39,19		17				21,22
BB35		39	25,7,26,6, 18,19		34				21,23
BB36		39	25,7,26,6, 18,19		16				21,22
BB37		18,39	25,7,26,6,19		37				21,23
BB38		18,39	25,7,26,6,19		15				21,22
BB39		19	25,7,26,6, 18,39		20				21,23
BB40		19	25,7,26,6, 18,39		36				21,22

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* TABLE XXVII. Propagation delay time (t_{PHL}) test - Continued.

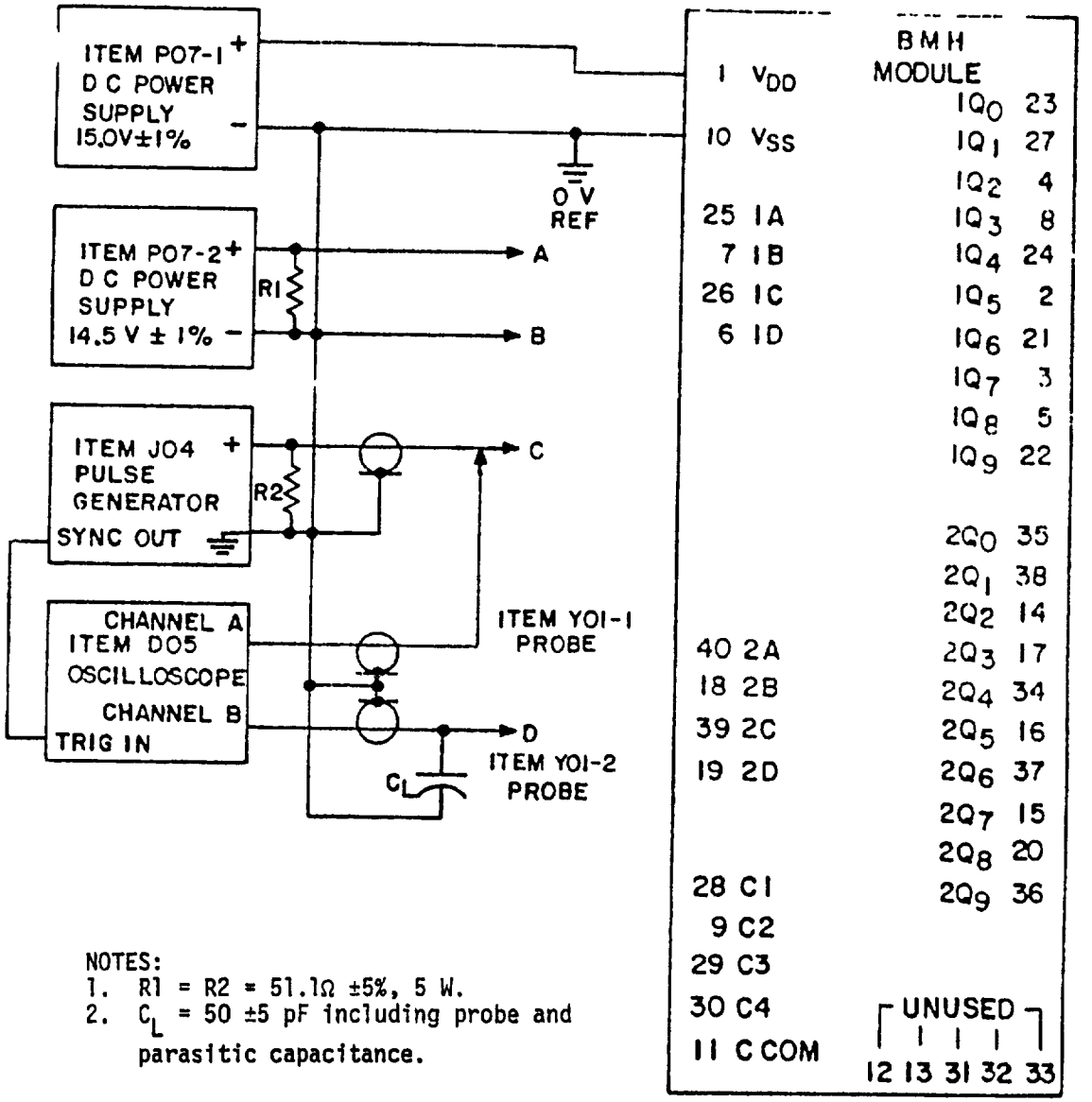
Test condition	Parameter	Connect				Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
BB41	Propagation delay time (t_{PHL}), V_{DD} = 15 V, C_L = 50 pF	Open	25,7,26,6, 40,39,19	18	35	14	140	ns	21,23
BB42		40	25,7,26,6, 39,19		38				21,23
BB43		Open	25,7,26,6, 40,39,19		14				21,22
BB44		40	25,7,26,6, 39,19		17				21,22
BB45		39	25,7,26,6, 40,19		34				21,23
BB46		39,40	25,7,26,6,19		16				21,23
BB47		39	25,7,26,6, 40,19		37				21,22
BB48		39,40	25,7,26,6,19		15				21,22
BB49		Open	25,7,26,6, 40,18,19	39	35				21,23
BB50		40	25,7,26,6, 18,19		38				
BB51		18	25,7,26,6, 40,19		14				
BB52		18,40	25,7,26,6,19		17				
BB53		Open	25,7,26,6, 40,18,19		34				21,22
BB54		40	25,7,26,6, 18,19		16				21,22

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* TABLE XXVII. Propagation delay time (t_{PHL}) test - Continued.

Test condition	Parameter	Connect				Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
BB55	Propagation delay time (t_{PHL}), V_{DD} = 15 V, C_L = 50 pF	18	25,7,26,6, 40,19	39	37	14	140	ns	21,22
BB56	↓	18,40	25,7,26,6,19	↓	15	↓	↓	↓	21,22
BB57	↓	Open	25,7,26,6, 40,18,39	19	35	↓	↓	↓	21,23
BB58	↓	40	25,7,26,6, 18,39	↓	36	↓	↓	↓	21,23
BB59	↓	Open	25,7,26,6, 40,18,39	↓	20	↓	↓	↓	21,22
BB60	↓	40	25,7,26,6, 18,39	↓	36	↓	↓	↓	21,22

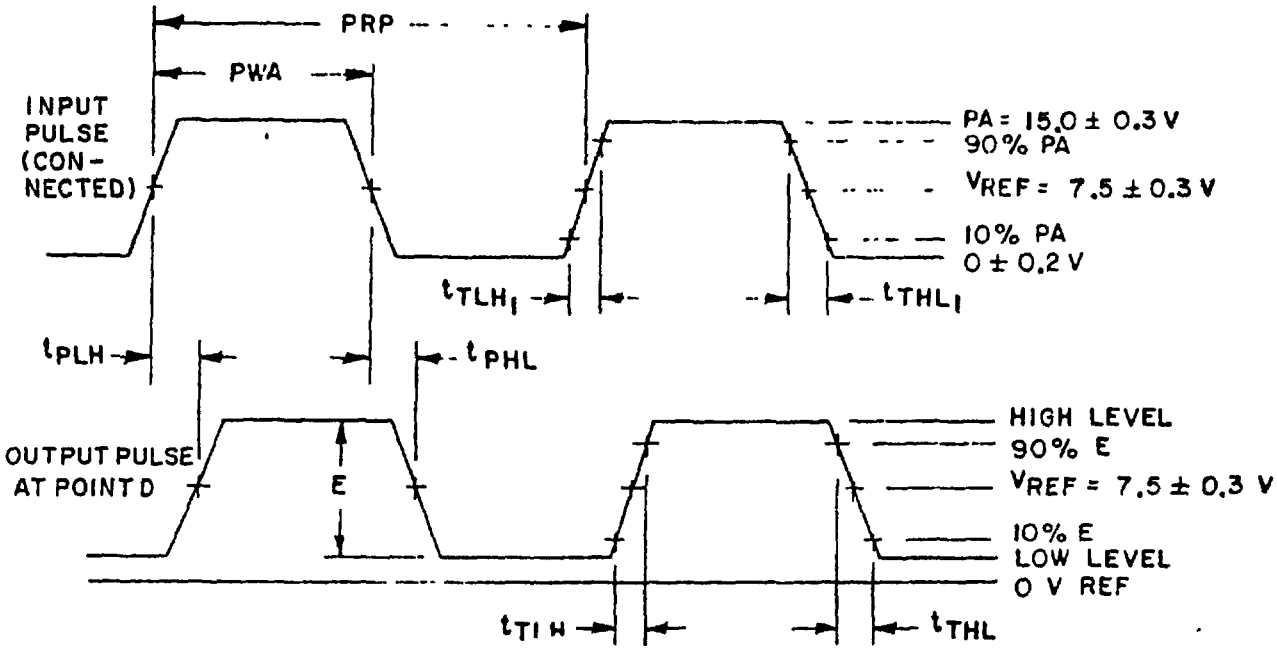
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NOTES:
1. R1 = R2 = 51.1Ω ±5%, 5 W.
2. C_L = 50 ±5 pF including probe and parasitic capacitance.

FIGURE 21. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test circuit for tables XXIV, XXV, XXVI and XXVII.

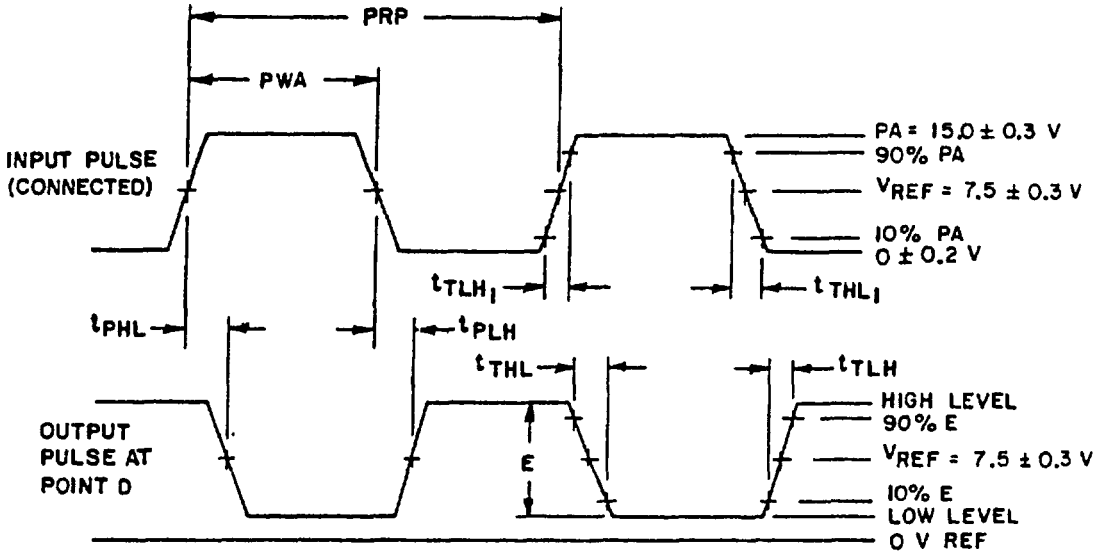
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NOTES:
1. Input pulse parameters:
   $t_{TLH_1} = t_{THL_1} = 10 \pm 2$ ns.
   PWA = 2.5 ±0.2 µs.
   PRP = 5.0 ±0.5 µs.
2. Before making the measurements specified in tables XXIV, XXV, XXVI and XXVII verify that the output pulse high level is $\geq$ 13.5 V and the low level is $\leq$ 1.5 V.
3. The lead length of electrical conductors between terminator resistor R1, R2 and input pins, probes YO1-1 and YO1-2 to input and output pins and between load circuit and output pin shall be less than 1.00 inch (25.4 mm).
4. All voltages are referenced to V_{SS}.

FIGURE 22. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test circuit waveforms for tables XXIV, XXV, XXVI and XXVII.

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NOTES:
1. Input pulse parameters:
   $t_{TLH_1} = t_{THL_1} = 10 \pm 2$ ns.
   PWA = 2.5 ±0.2 µs.
   PRP = 5.0 ±0.5 µs.
2. Before making the measurements specified in tables XXIV, XXV, XXVI and XXVII verify that the output pulse high level is ≥ 13.5 V and the low level is ≤ 1.5 V.
3. The lead length of electrical conductors between terminator resistor R1, R2 and input pins, probes Y01-1 and Y01-2 to input and output pins and between load circuit and output pins shall be less than 1.00 inch (25.4 mm).
4. All voltages are referenced to V_{SS}.

FIGURE 23. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test circuit waveforms for tables XXIV, XXV, XXVI and XXVII.

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TABLE XXVIII. Capacitor test.

Test condition	Parameter		Connect A to pin	Connect B to pin	Connect C to pin	Initial limits 25°±5°C ambient Min	Initial limits 25°±5°C ambient Max	Unit	Test figure
CC1	C1	Capacitor	28	Open	11	176	264	pF	24
CC2	C2	↓	9	↓	↓	↓	↓	↓	↓
CC3	C3		29						
CC4	C4		30						

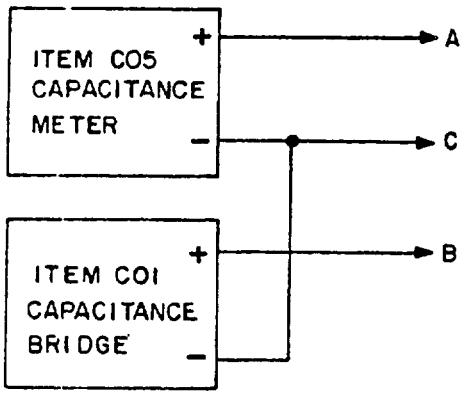
TABLE XXIX. Power supply filter capacitance test.

Test condition	Parameter	Connect A to pin	Connect B to pin	Connect C to pin	Initial limits 25°±5°C ambient Min	Initial limits 25°±5°C ambient Max	Unit	Test figure
DD	Power supply filter capacitance	Open	1	10	0.007	0.013	µF	24

TABLE XXX. Input capacitance test.

Test condition	Parameter	Connect A to pin	Connect B to pin	Connect C to pin	Initial limits 25°±5°C ambient Min	Initial limits 25°±5°C ambient Max	Unit	Test figure
EE1	Input capacitance	25	Open	10	2	15	pF	24
EE2	↓	7	↓	↓	↓	↓	↓	↓
EE3		26						
FE4		6						
EE5		40						
EE6		18						
EE7		39						
EE8		19						

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		BMH MODULE	
1	V_{DD}	$1Q_0$	23
10	V_{SS}	$1Q_1$	27
		$1Q_2$	4
25	1A	$1Q_3$	8
7	1B	$1Q_4$	24
26	1C	$1Q_5$	2
6	1D	$1Q_6$	21
		$1Q_7$	3
		$1Q_8$	5
		$1Q_9$	22
		$2Q_0$	35
		$2Q_1$	38
		$2Q_2$	14
40	2A	$2Q_3$	17
18	2B	$2Q_4$	34
39	2C	$2Q_5$	16
19	2D	$2Q_6$	37
		$2Q_7$	15
		$2Q_8$	20
28	C1	$2Q_9$	36
9	C2		
29	C3		
30	C4	UNUSED	
11	C COM	12 13 31 32 33	

NOTES:
1. Null out the effects of the test leads prior to connecting the module.
2. Do not attach test connector or wires to any of the module pins other than the pins under test. Stray capacitance added to other pins may affect the readings.
3. Capacitance bridge C01 setup:
   Frequency = 1.0 kHz.
   V_{pk} = 0.3 V maximum.

FIGURE 24. Capacitor, power supply filter and input capacitance test circuit for tables XXVIII, XXIX and XXX.

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